

Name: _____

GCSE Physics Unit 1 Foundation Tier

Once you have completed a paper and have marked it, fill out the table below, using the grade boundaries provided to add your grade. Then list which questions you did well and which questions you struggled with. For the questions you struggled with, review why and at the reasons. The most common reasons are

- Not understanding basic concepts (e.g. simple definitions)
- Calculation errors (forgetting to square numbers, etc.)
- Not reading questions carefully
- Not answering the full question (describe vs. explain)
- Laboratory/practical setup mistakes

For these questions it is also useful add notes to the question to help you next time and take a photo of these questions. Use these then to guide further revision.

Year	Score /80	Grade	Good Questions	Bad Questions	Reasons
2017					
2018					
2019					
2022					
2023					

Grade boundaries - Raw Score

Year	G	F	E	D	C	Max UMS
2017	17	21	25	30	35	40
2018	19	23	27	32	37	42
2019	19	23	27	31	36	41
2022	13	18	23	28	33	38
2023	11	18	25	32	40	48

Surname	Centre Number	Candidate Number
Other Names		0



GCSE – NEW

3420U10-1



**PHYSICS – Unit 1:
Electricity, Energy and Waves**

FOUNDATION TIER

MONDAY, 19 JUNE 2017 – MORNING

1 hour 45 minutes

For Examiner's use only		
Question	Maximum Mark	Mark Awarded
1.	8	
2.	9	
3.	15	
4.	7	
5.	13	
6.	8	
7.	8	
8.	12	
Total	80	

ADDITIONAL MATERIALS

In addition to this paper you may require a calculator, a ruler and a drawing compass.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen.
Write your name, centre number and candidate number in the spaces at the top of this page.
Answer **all** questions.
Write your answers in the spaces provided in this booklet.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.
Question **6(a)** is a quality of extended response (QER) question where your writing skills will be assessed.

Equations

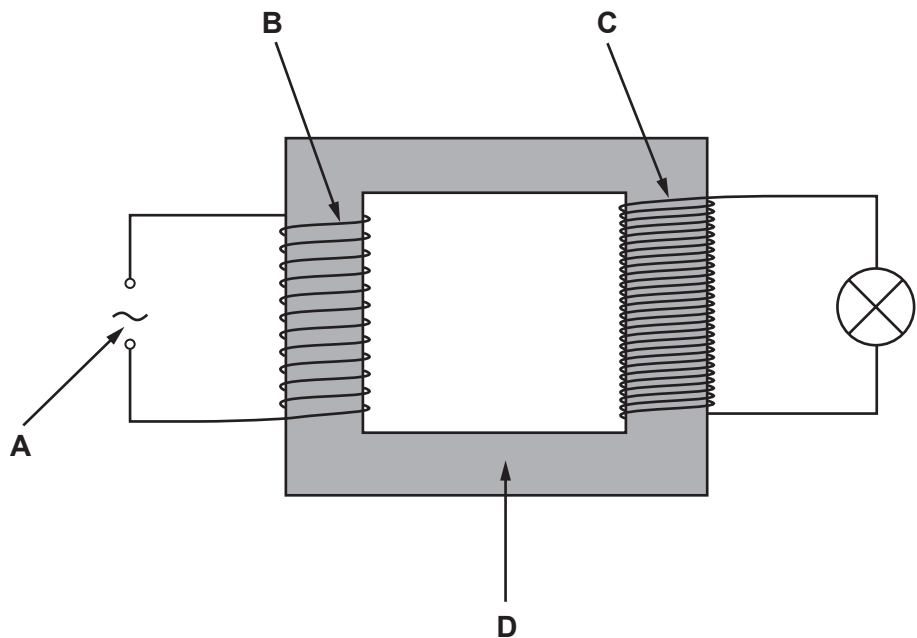
current = $\frac{\text{voltage}}{\text{resistance}}$	$I = \frac{V}{R}$
total resistance in a series circuit	$R = R_1 + R_2$
energy transferred = power $\times$ time	$E = Pt$
power = voltage $\times$ current	$P = VI$
% efficiency = $\frac{\text{energy [or power] usefully transferred}}{\text{total energy [or power] supplied}} \times 100$	
density = $\frac{\text{mass}}{\text{volume}}$	$\rho = \frac{m}{V}$
units used (kWh) = power (kW) $\times$ time (h) cost = units used $\times$ cost per unit	
wave speed = wavelength $\times$ frequency	$v = \lambda f$
speed = $\frac{\text{distance}}{\text{time}}$	
pressure = $\frac{\text{force}}{\text{area}}$	$p = \frac{F}{A}$
change in thermal energy = mass $\times$ specific heat capacity $\times$ change in temperature	$\Delta Q = mc\Delta\theta$
thermal energy for a change of state = mass $\times$ specific latent heat	$Q = mL$
V_1 = voltage across the primary coil V_2 = voltage across the secondary coil N_1 = number of turns on the primary coil N_2 = number of turns on the secondary coil	$\frac{V_1}{V_2} = \frac{N_1}{N_2}$

SI multipliers

Prefix	Multiplier
m	1×10^{-3}
k	1×10^3
M	1×10^6

Answer **all** questions.

1. The diagram shows a transformer that is used in a laboratory experiment.



(a) Draw one line from each letter to the correct label.

[3]

A	primary coil
B	laminated iron core
C	secondary coil
D	a.c. input (supply)

(b) The diagram shows a step-up transformer. Complete the sentences by underlining the correct word or phrase in the brackets. [3]

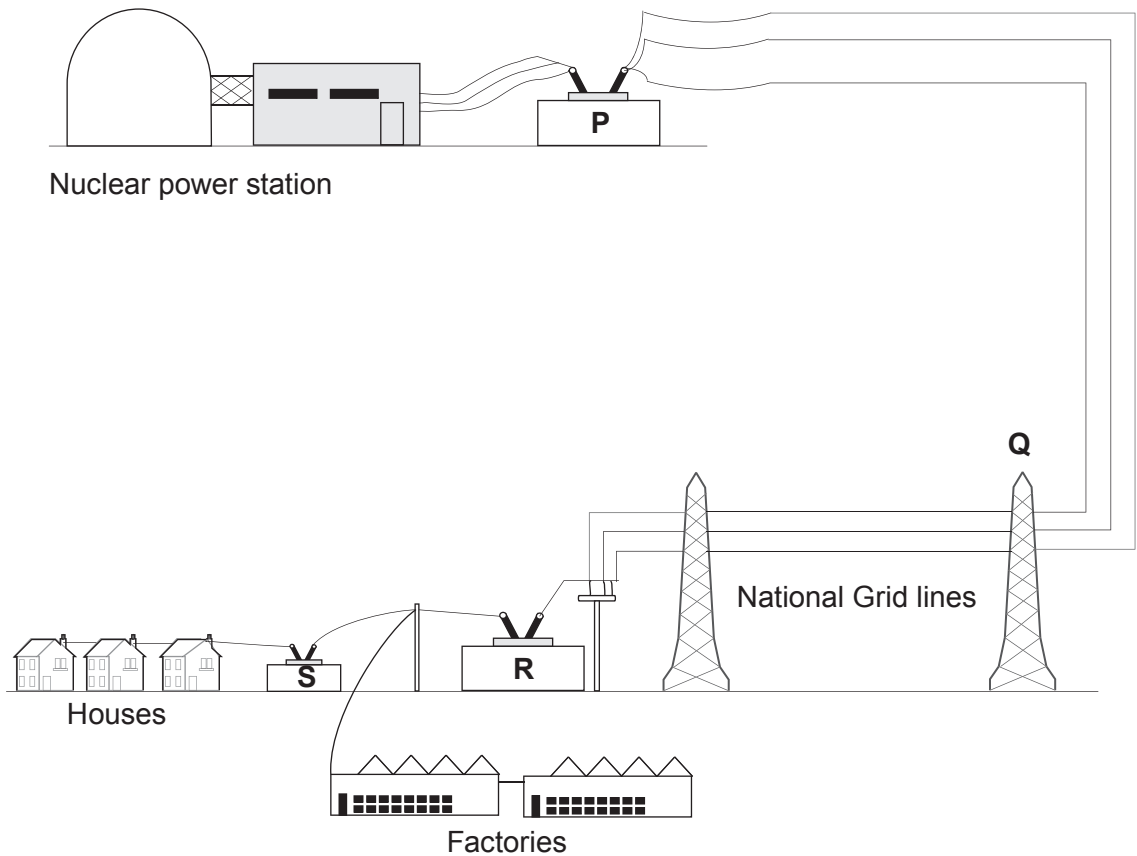
A step-up transformer that is 100% efficient will:

- make the output voltage (**smaller / stay the same / bigger**).
- make the output current (**smaller / stay the same / bigger**).
- make the output power (**smaller / stay the same / bigger**).

(c) The diagram below shows part of the National Grid. Power stations are used to generate electricity. They are linked to houses and factories by a network of cables.

(i) Which letter **P**, **Q**, **R** or **S** shows a step-up transformer?

[1]

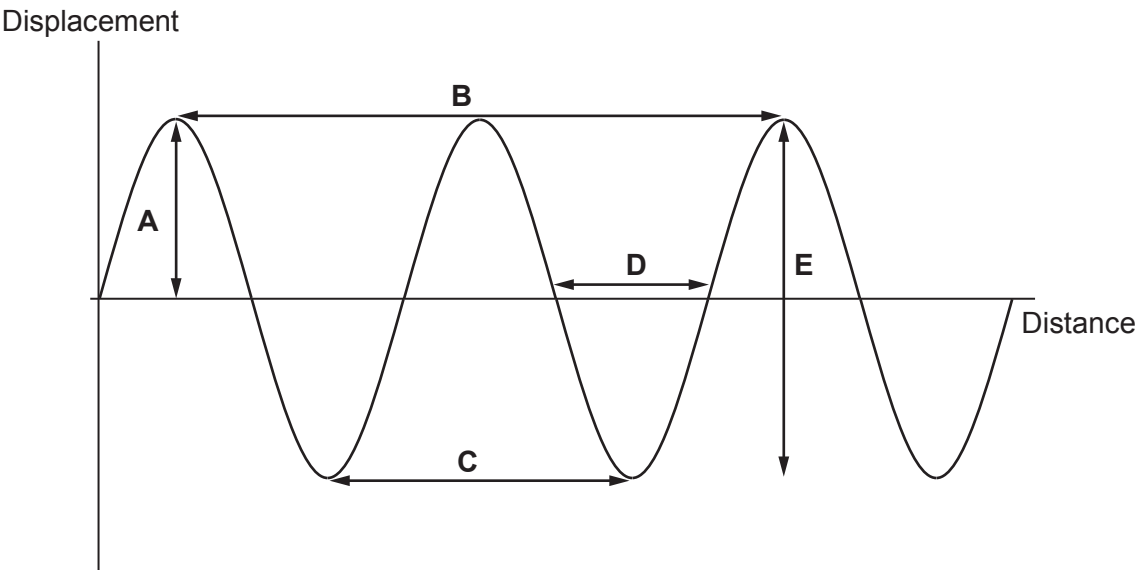


(ii) State **one** environmental advantage of using a nuclear power station compared to a gas-fired power station. [1]

.....

.....

2. The diagram represents water waves on the surface of a swimming pool.



- (a) How many complete waves are shown in the diagram? [1]
- (b) Which letter **A**, **B**, **C**, **D** or **E** shows the amplitude of the wave? [1]
- (c) Select a word from the box to complete the following sentences.

frequency	time	speed	wavelength	amplitude
-----------	------	-------	------------	-----------

- (i) The number of waves per second is the [1]
- (ii) The maximum displacement of the wave is the [1]
- (d) A wave peak takes 0.5s to travel the distance labelled **B** on the diagram. The speed of the wave is 20 cm/s.

(i) Use the equation:

$$\text{distance} = \text{speed} \times \text{time}$$

to calculate the distance travelled by the wave. [2]

Distance = cm

- (ii) Calculate the wavelength. [1]

Wavelength = cm



(e) Water waves are transverse waves.

(i) Give another example of a transverse wave.

[1]

.....

(ii) Sound waves are not transverse waves. What type of waves are they?

[1]

.....

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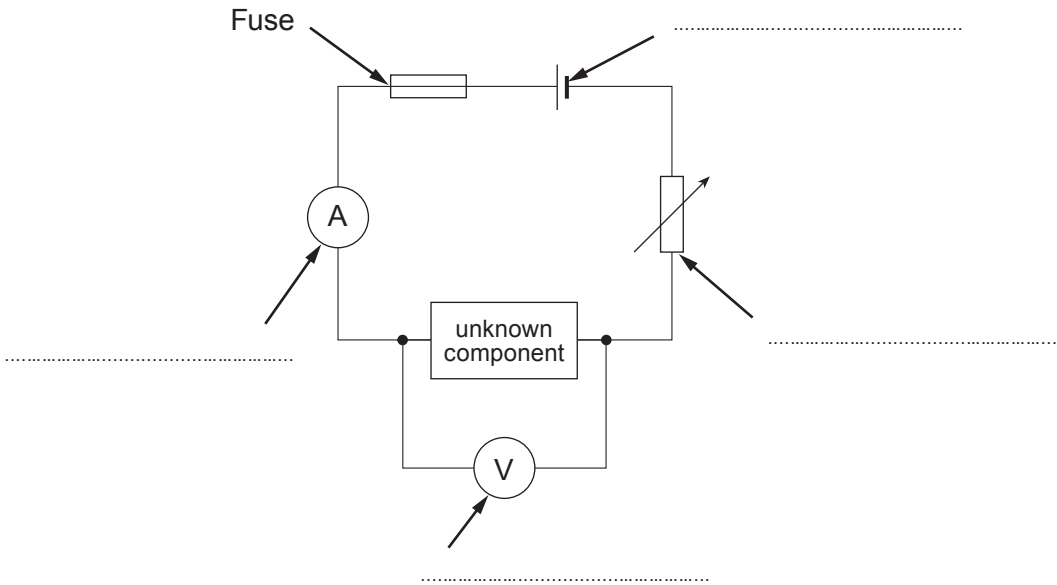
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Examiner
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3. A student is given an unknown electrical component in a sealed box. He carries out an experiment to identify it.

He sets up the circuit below to investigate how the current changes with voltage for the unknown component.



(a) **Complete the labelling** on the circuit diagram. [4]

(b) The data shown in the table is collected from the experiment.

Voltage (V)	−0.4	−0.2	0.0	0.2	0.4	0.6	0.8	1.0
Current (mA)	0	0	0	0	0	5	20	50

(i) To prevent the unknown component being damaged the current through it **must not** be greater than 150 mA.

Circle the appropriate fuse that should be used in the circuit. [1]

10 mA 100 mA 200 mA 500 mA

(ii) (I) State the current in **amps (A)** when the voltage is 1.0 V. [1]

Current = A

(II) Use the equation:

[2]

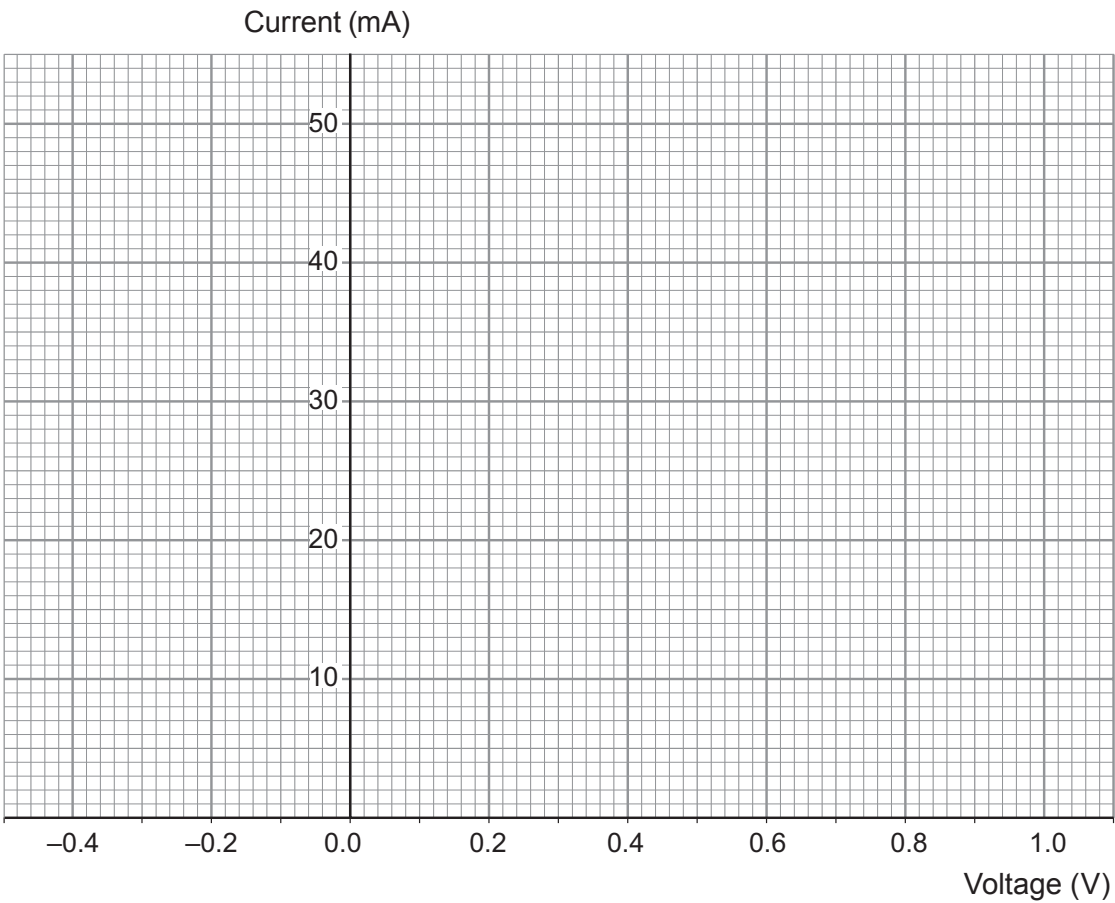
resistance = $\frac{\text{voltage}}{\text{current}}$

to calculate the resistance of the unknown component at 1.0 V.

Resistance = Ω

(iii) Plot the data in the table on the grid below and draw a suitable line.

[3]



(iv) Use the graph to find the current when the voltage is 0.7 V.

[1]

Current = mA

(v) Describe the relationship between the current and the voltage in the range 0.4 V to 1.0 V.

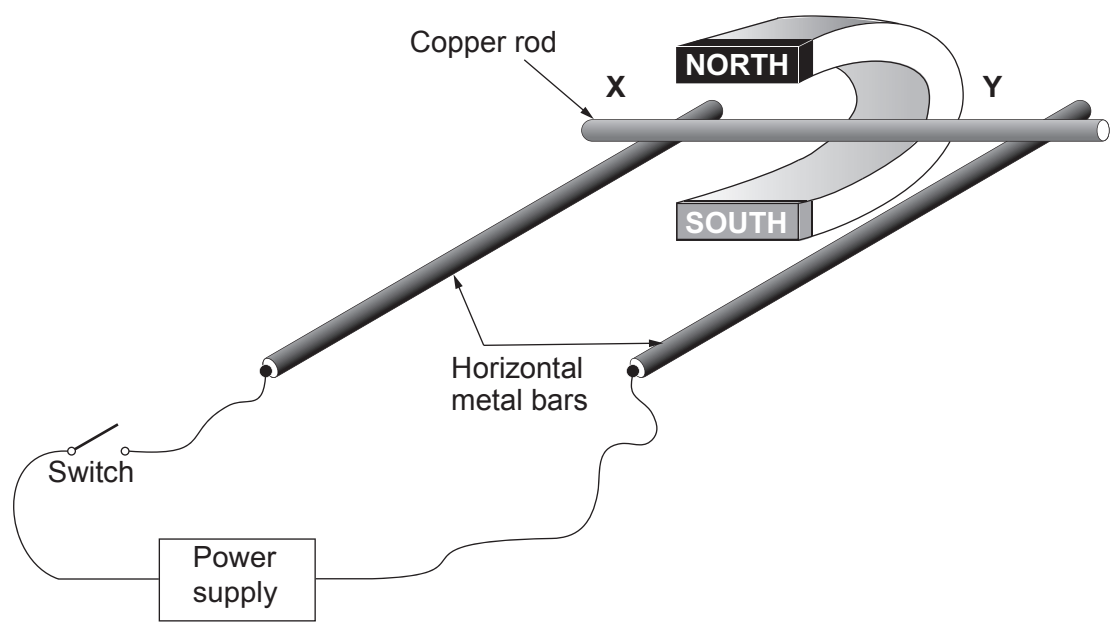
[2]

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.....
.....

(vi) Identify the unknown component.

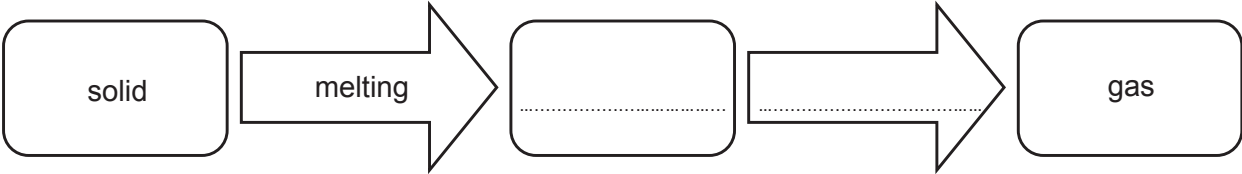
[1]

4. The following experiment is set up in a school laboratory for students.



- (a) (i) Explain why, when the switch in the circuit is closed, the copper rod starts to roll along the two horizontal metal bars. [2]
-
-
-
- (ii) When the switch is closed the copper rod moves away from the magnet. One of the students concludes that the direction of the current is from **X** to **Y**. Explain if you agree or disagree with their conclusion. [2]
-
-
-
- (iii) State **two** changes that could be made to the apparatus so that the copper rod moves faster when the switch is closed. [2]
1.
2.
- (b) The effect observed in this experiment is used in many everyday devices. Give an example of **one** such use. [1]
-

5. The diagram gives some information about what happens when a solid is heated.

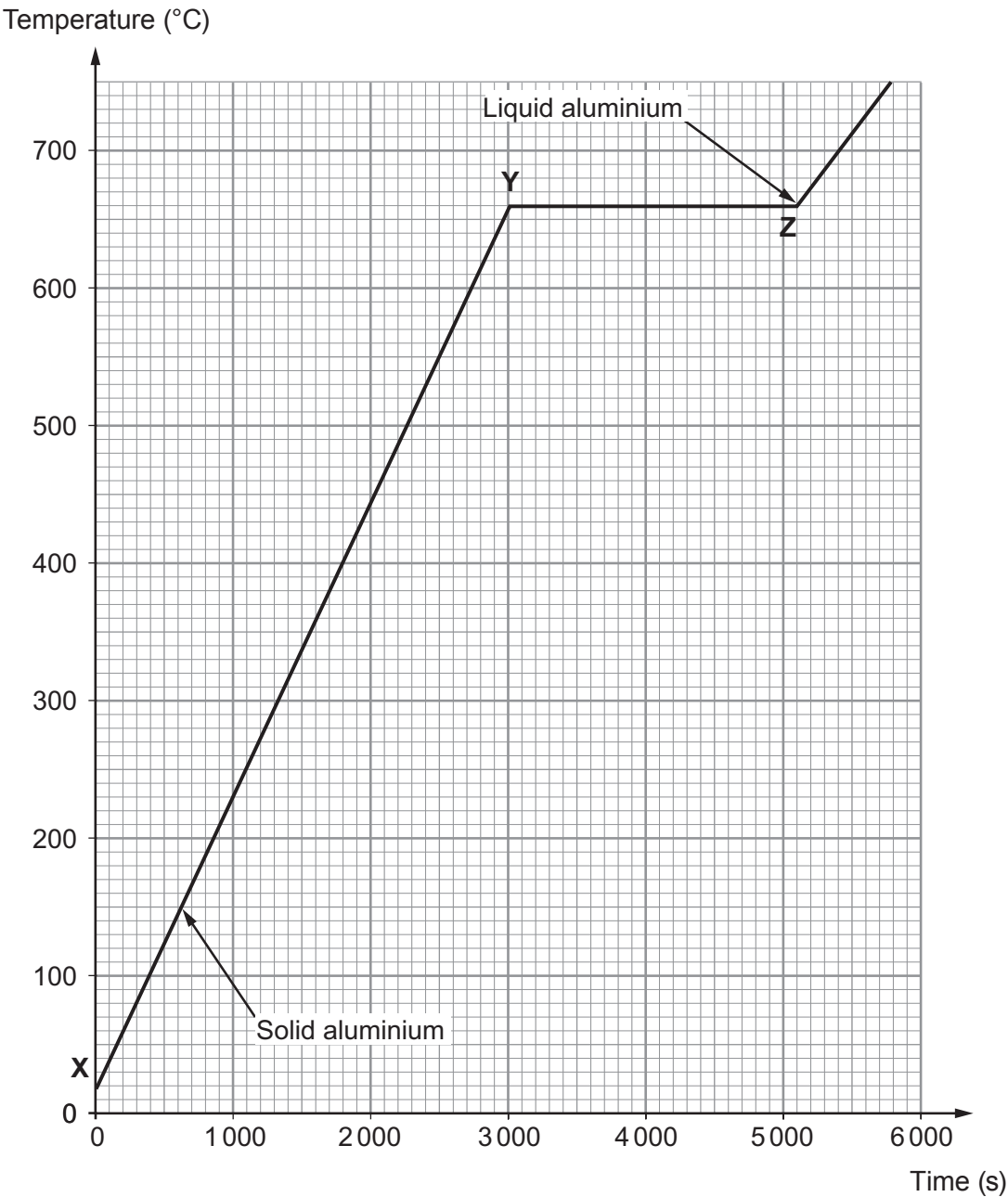


(a) Complete the diagram. [2]

(b) Tick (✓) the **two** correct statements about a solid. [2]

A solid has the lowest density of the three states of matter.	
The atoms in a solid are in fixed positions.	
The atoms in a solid transfer heat by convection.	
A solid is always a good conductor.	
A solid has atoms that vibrate more as they gain energy.	

- (c) Aluminium cans are frequently recycled. The aluminium cans are collected by local councils as part of household waste. They are sent to a furnace where the cans are heated to melt them. The aluminium is then cooled so that it can be reused for the manufacture of other items. The graph shows how the temperature of the aluminium cans in the furnace changes with time.



- (i) State the temperature at which the aluminium cans change from a solid into a liquid. [1]

Temperature = °C

- (ii) State the time it takes to heat the aluminium cans to their melting point. [1]

Time = s

- (d) The heat transferred during the heating process, between points **X** and **Y** is 288 000 000 J.

- (i) Use your answer from (c)(ii) and the equation:

$$\text{power} = \frac{\text{heat transferred}}{\text{time}}$$

to calculate the power **in kW** of the heater in the furnace.

[3]

Power = kW

- (ii) The temperature change of the aluminium cans during the heating process **X** to **Y** is 640 °C.

Use information given above and the equation:

$$\text{mass} = \frac{\text{heat transfer}}{(\text{specific heat capacity} \times \text{temperature change})}$$

to calculate the mass of aluminium cans that were heated in the furnace.
(Specific heat capacity of aluminium = 900 J/kg °C)

[2]

Mass = kg

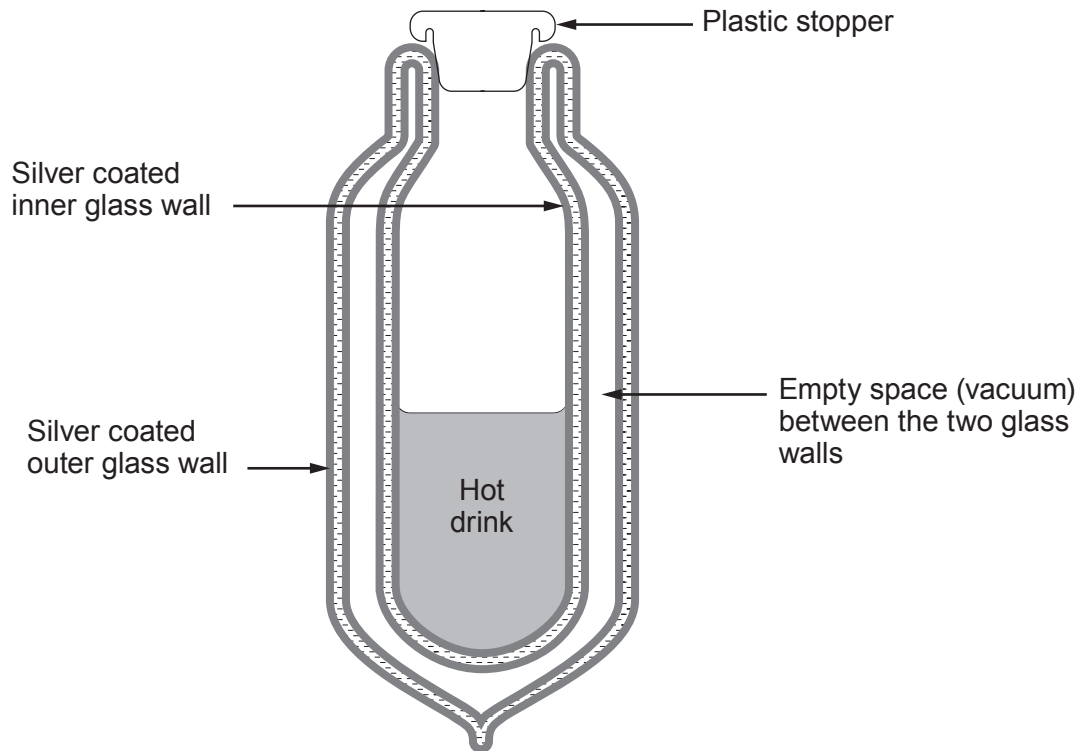
- (e) Use an equation from page 2 to calculate the heat transfer required to melt 1 500 kg of aluminium cans from solid to liquid at its melting point.
(Specific latent heat of fusion of aluminium, $L = 400\,000 \text{ J/kg}$)

[2]

Heat transfer = J

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6. On a cold day a vacuum flask can be used to keep drinks hot for many hours. The diagram shows a vacuum flask with its main features labelled.



- (a) Explain in terms of conduction, convection and radiation how the labelled features of the flask help to keep the drink hot for such a long time. [6 QER]

- (b) A drink at 90 °C is poured into the flask. After 2 hours the temperature of the drink is 80 °C. After 10 hours the temperature of the drink is 60 °C.

The company who make the vacuum flask claim that a hot drink contained in the flask will cool down by 5 °C every hour. Explain if the claim made by the company is always correct. [2]

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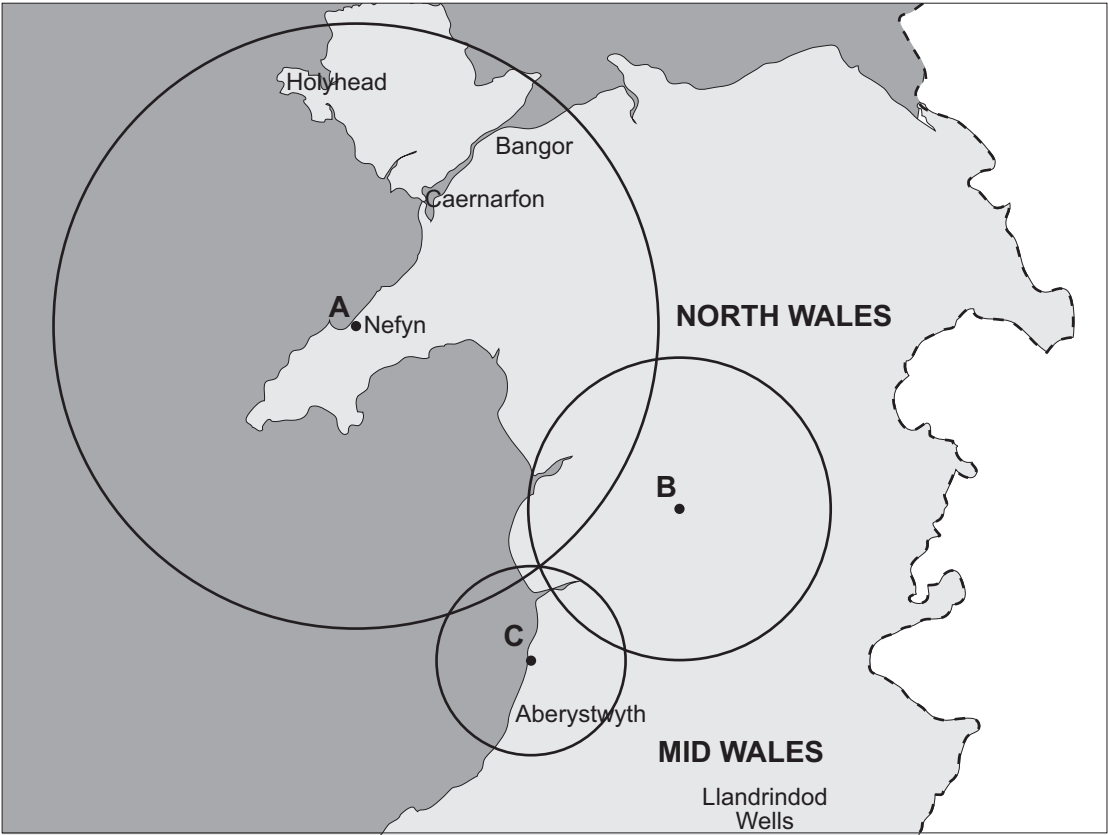
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7. The waves from an earthquake can be monitored by seismological stations located on the surface of the Earth.



The diagram shows circles drawn around three seismological stations **A**, **B** and **C** in Mid and North Wales. The radius of each circle shows the distance of the epicentre from each station.

- (a) (i) **Label on the diagram** with a cross (X) the location of the epicentre. [1]
- (ii) Explain why at least three seismological stations are needed to locate the epicentre. [2]

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- (b) Explain which station (**A**, **B** or **C**) detected the earthquake first. [2]

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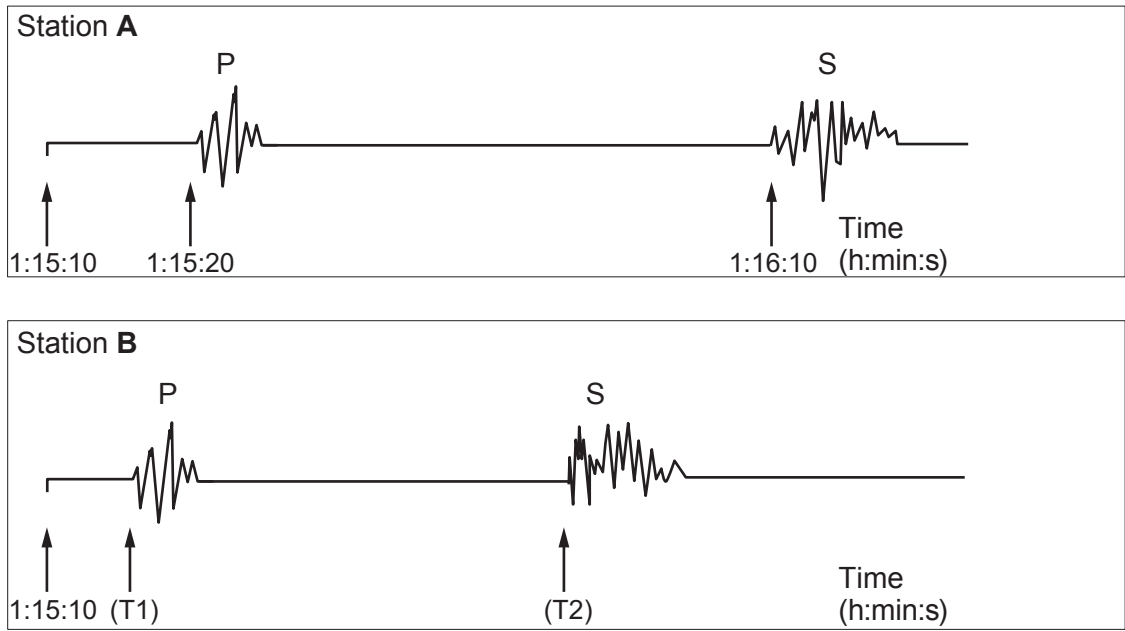
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- (c) The diameter of the circle around station A **is double** the diameter of the circle around station B.

The earthquake happened at 1 hour : 15 minutes : 10 seconds (1:15:10) in the morning.

The following signals were obtained at stations **A** and **B**.



Due to an error on the equipment the **arrival times** (T1 and T2) for the two signals were not recorded at station **B**.

Using signal information from station **A**, determine the times that the P waves and the S waves arrived at station **B**. Explain how you arrived at your answers. [3]

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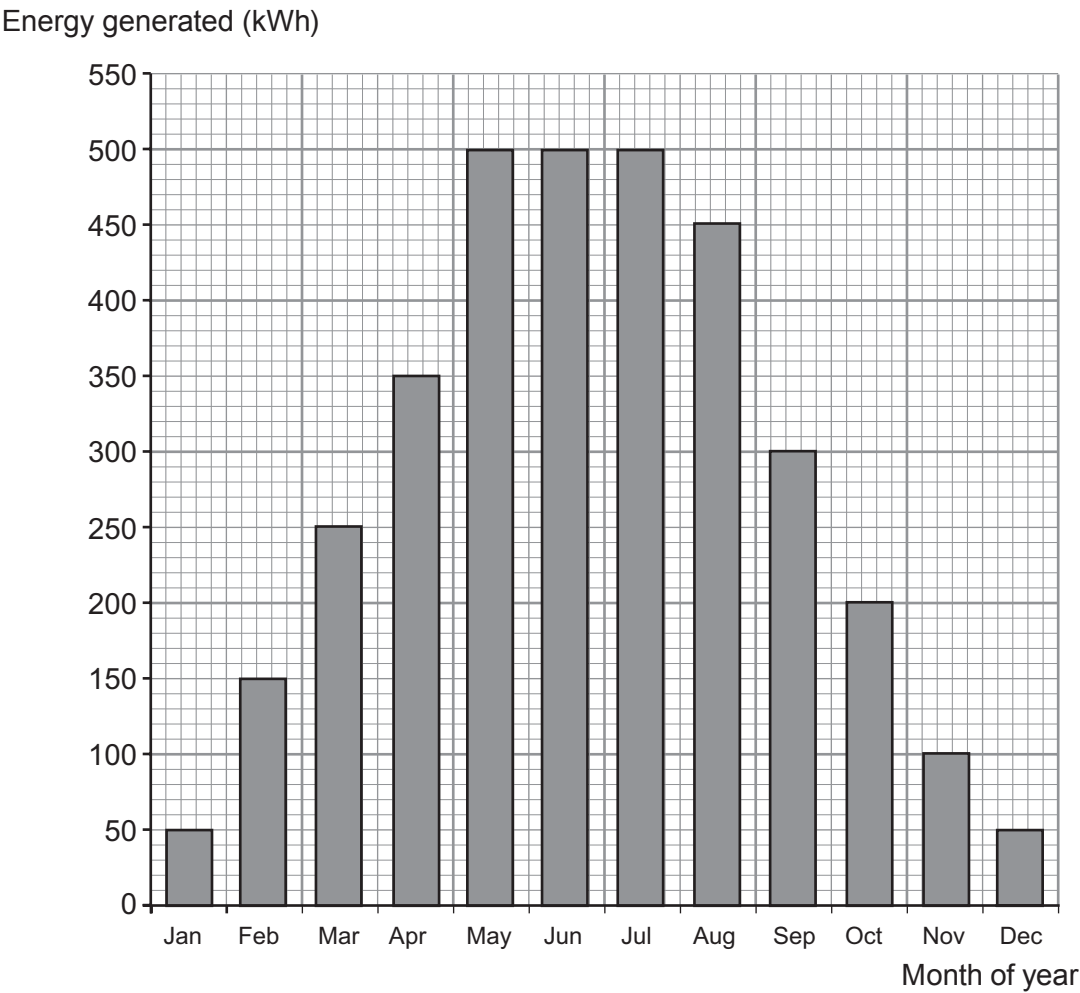
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Arrival time of P waves (T1) = hours minutes seconds.

Arrival time of S waves (T2) = hours minutes seconds.

8. A family from Wales researched photovoltaic (PV) cells before they bought some. They were told by the manufacturer that they could expect the PV cells to generate 3400 kWh of electricity per year. **Figure 1** shows the energy generated in kWh from PV cells in a typical year in Wales. This was included in an advert for the PV cells.

Figure 1



- (a) Use the information from **Figure 1** to validate the claim that PV cells could generate 3400 kWh annually. [2]

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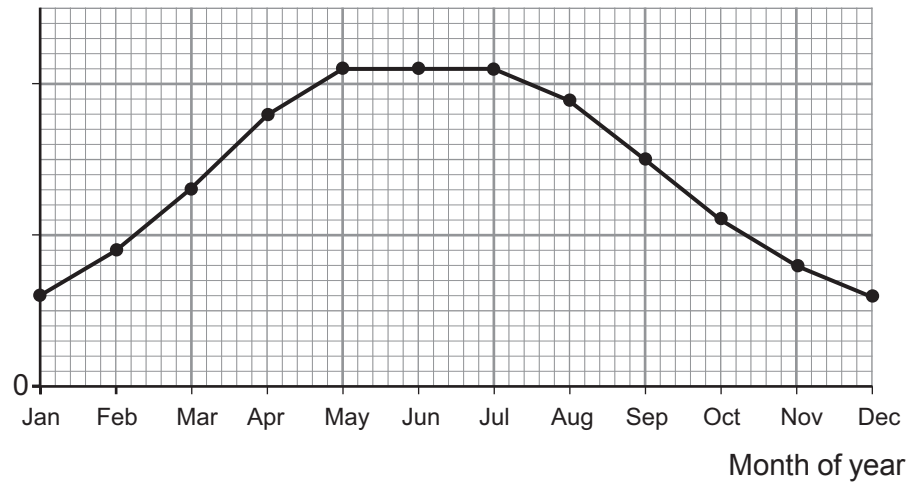
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- (b) Sunshine data for Wales over the last 30 years was obtained from the Welsh MET weather office. **Figure 2** shows the mean number of sunshine hours for each month in Wales.

Figure 2

Mean number of
sunshine hours



- (i) Describe how the information from **Figure 2** supports the general trend shown in **Figure 1**. [1]

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- (ii) The Welsh MET weather office also had data for the **maximum** number of sunshine hours for each month. State why the manufacturers were **not** allowed to use this data in their advert. [1]

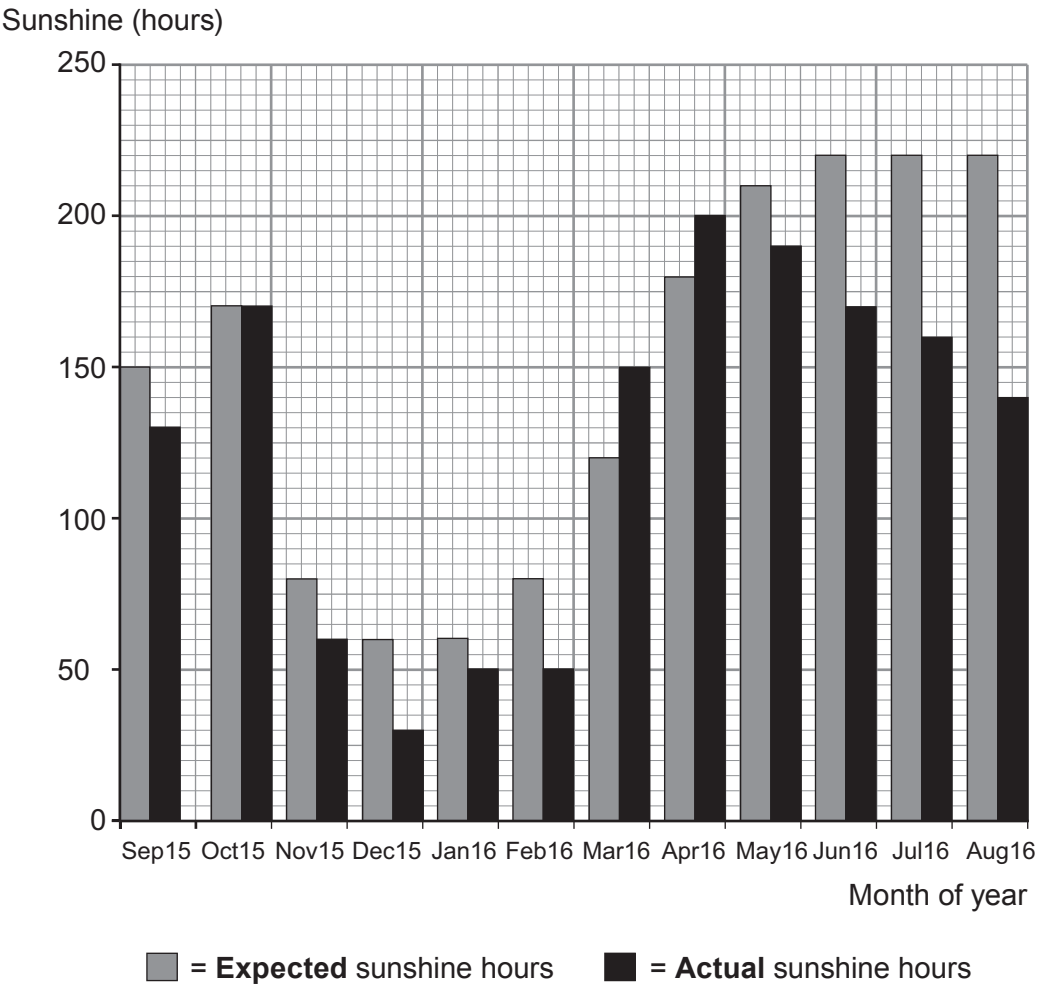
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- (c) In September 2015 the family had PV cells installed on their south facing roof. They collected 12 months of actual data from their PV cells. **Figure 3** compares the **expected** sunshine hours with **actual** sunshine hours. The **expected** sunshine hours data were obtained from local weather station records.

Figure 3



- (i) Use information from **Figure 3** to tick (✓) the **three** correct statements. [3]

In Dec15 the **actual** number of sunshine hours was double the **expected** number.

☐

Apr 16 was the month that had the most **actual** sunshine hours.

☐

June, July and August have different **expected** sunshine hours.

☐

Jan 16 had the least number of **actual** sunshine hours.

☐

Apr 16 had 4 times the **actual** number of sunshine hours compared to Jan 16.

☐

There is only one month where the **actual** and the **expected** sunshine hours were the same.

☐

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- (ii) In April 16 the PV cells generated 600 kWh. The manufacturers claim the PV cells will produce 3 kW in sunshine. Use an equation from page 2 and **Figure 3** to validate this claim. [2]

.....

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.....

- (d) The PV cells were expected to produce 3 400 kWh of electricity. However, the total number produced was 3 670 kWh.

- (i) The family saves 29 p for each kWh of energy generated. Calculate how much extra money they saved. [2]

Extra savings = p

- (ii) If the PV cells continue to generate more energy than expected what effect would this have on their payback time? [1]

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END OF PAPER

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Surname	Centre Number	Candidate Number
Other Names		0



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PHYSICS – Unit 1:
Electricity, Energy and Waves

FOUNDATION TIER

FRIDAY, 15 JUNE 2018 – MORNING

1 hour 45 minutes

For Examiner’s use only		
Question	Maximum Mark	Mark Awarded
1.	9	
2.	9	
3.	13	
4.	7	
5.	7	
6.	10	
7.	12	
8.	13	
Total	80	

ADDITIONAL MATERIALS

In addition to this paper you will require a calculator, a ruler and a drawing compass.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen. Do not use correction fluid.
Write your name, centre number and candidate number in the spaces at the top of this page.
Answer **all** questions.
Write your answers in the spaces provided in this booklet. If you run out of space use the additional page at the back of the booklet.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.
The assessment of the quality of extended response (QER) will take place in question **3(a)**.



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Equations

current = $\frac{\text{voltage}}{\text{resistance}}$	$I = \frac{V}{R}$
total resistance in a series circuit	$R = R_1 + R_2$
energy transferred = power $\times$ time	$E = Pt$
power = voltage $\times$ current	$P = VI$
% efficiency = $\frac{\text{energy [or power] usefully transferred}}{\text{total energy [or power] supplied}} \times 100$	
density = $\frac{\text{mass}}{\text{volume}}$	$\rho = \frac{m}{V}$
units used (kWh) = power (kW) $\times$ time (h) cost = units used $\times$ cost per unit	
wave speed = wavelength $\times$ frequency	$v = \lambda f$
speed = $\frac{\text{distance}}{\text{time}}$	
pressure = $\frac{\text{force}}{\text{area}}$	$p = \frac{F}{A}$
change in thermal energy = mass $\times$ specific heat capacity $\times$ change in temperature	$\Delta Q = mc\Delta\theta$
thermal energy for a change of state = mass $\times$ specific latent heat	$Q = mL$
V_1 = voltage across the primary coil V_2 = voltage across the secondary coil N_1 = number of turns on the primary coil N_2 = number of turns on the secondary coil	$\frac{V_1}{V_2} = \frac{N_1}{N_2}$

SI multipliers

Prefix	Multiplier
m	1×10^{-3}
k	1×10^3
M	1×10^6



Answer all questions.

1. (a) The diagram below shows **part** of the electromagnetic (em) spectrum.

Microwaves	Infra-red	Visible light	Ultraviolet
------------	-----------	---------------	-------------

Use **only** the regions of the em spectrum **shown in the diagram** to answer the following questions.

(i) Name the region of the em spectrum with the longest wavelength. [1]

.....

(ii) Name the region of the em spectrum with the lowest frequency. [1]

.....

(b) Name **one** region of the em spectrum not shown in the diagram in part (a). [1]

.....

(c) Waves can either be described as transverse or longitudinal. Sound waves are an example of longitudinal waves whereas visible light waves are transverse.
Tick (✓) the **two** correct statements below. [2]

Ultraviolet waves are longitudinal waves ☐

Longitudinal waves cannot be reflected ☐

Microwaves are transverse waves ☐

In a longitudinal wave the vibration of the particles is parallel to the direction of the wave ☐

Sound waves travel slowly in a vacuum ☐



(d) The table below gives information about the frequency and wavelength of sound waves in different materials.

Material	Frequency (Hz)	Wavelength (m)
air	170	2
water	170	9
iron	170	29

Use the information in the table to answer the questions below.

(i) Use the equation:

wave speed = frequency $\times$ wavelength

to calculate the speed of sound waves in air. [2]

Wave speed = m/s

(ii) The sound wave travels from air into water. Its frequency stays the same. Without further calculation explain whether its speed increases, decreases or stays the same. [2]

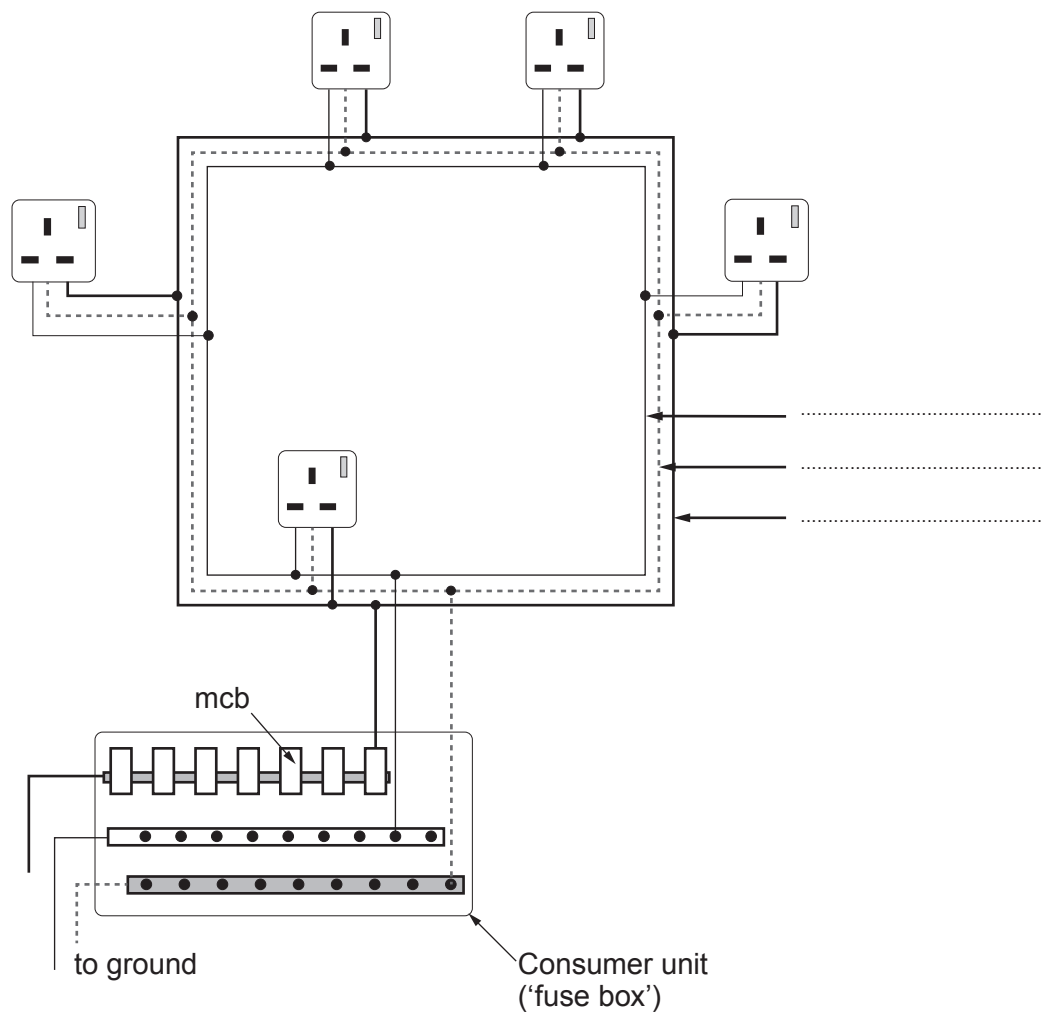
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2. The diagram below shows a ring main circuit for a house.



(a) (i) **Label** the earth, neutral and live wires on the diagram above. [2]

(ii) Underline a word in each bracket to correctly complete the sentences below. [3]

The ring main is a looped (**double / parallel / series**) circuit. The cables in the ring main can be made (**thinner / thicker / longer**) because there are 2 paths for the (**voltage / current / power**).



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(b) A 1.2 kW kettle is plugged into the ring main. It is used for 0.5 hours in a day.
Use equations from page 2 to answer the following questions.

(i) Calculate the number of units (kWh) the kettle uses each day. [2]

Units used = kWh

(ii) Calculate the cost of using the kettle each day if electricity costs 15p per unit. [2]

Cost = p

9



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3. Rhys and Elliot have a small piece of an unknown metal. The metal has an **irregular shape**. To identify the metal they find its density and compare the value to known values of the density of common metals.

(a) Describe a method they could use to find the density of the metal. [6 QER]

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(b) The table below gives data on the density of some common metals.

Metal	Density (g/cm ³)
aluminium	2.70
copper	8.96
gold	19.32
iron	7.87
tin	7.26

(i) Rhys and Elliot calculate the density of their metal to be 8.1 g/cm³.
State which metal the irregular shape is most likely to be. Give a reason for your
answer. [2]

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(ii) Rhys and Elliot are not confident that they can correctly identify the metal. Suggest
why they think this. [1]

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(iii) The table below shows their results.

Mass (g)	Volume (cm ³)	Density (g/cm ³)
65	8	8.1

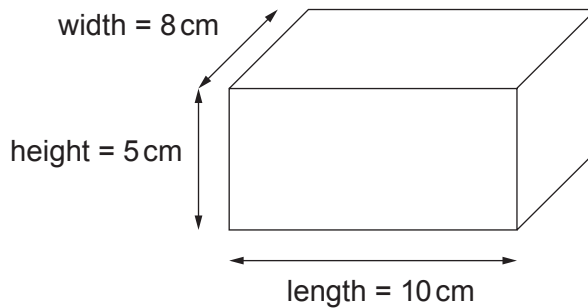
Suggest how Rhys and Elliot could get a more accurate value for the density. [1]

.....

.....



- (c) The boys notice that gold has a high density of 19.32 g/cm^3 and they are interested in the mass of a gold block.



- (i) Use the equation:

$$\text{volume} = \text{length} \times \text{height} \times \text{width}$$

to calculate the volume of the gold block shown above.

[1]

Volume = cm^3

- (ii) Use the equation:

$$\text{mass} = \text{density} \times \text{volume}$$

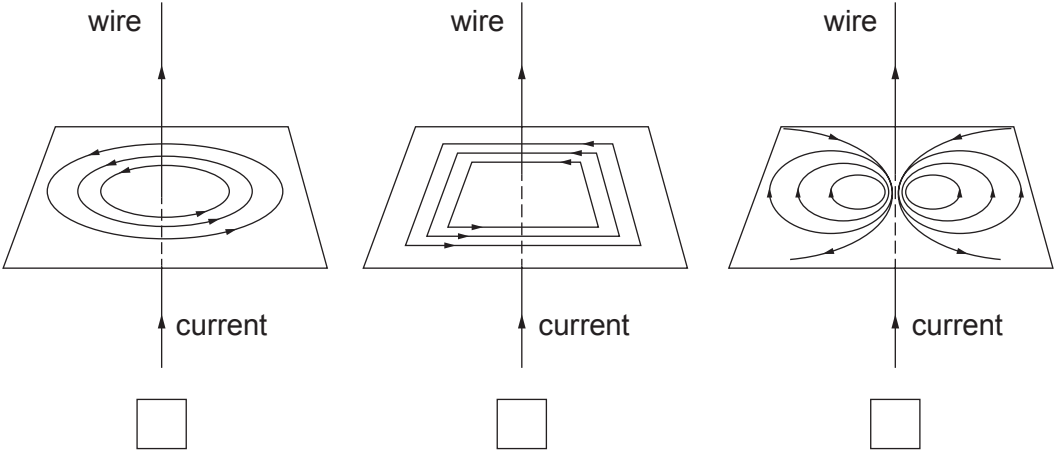
to calculate the mass of the gold block.

[2]

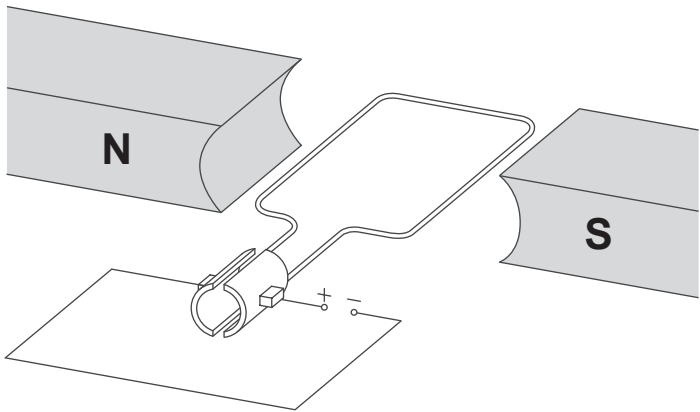
Mass = g



4. (a) Tick (✓) the box below the diagram which correctly shows the magnetic field pattern around a current-carrying straight wire. [1]



(b) The diagram below shows a simple electric motor. When there is a current in the coil it experiences a force due to the magnetic field and starts to spin.



(i) State **one** way in which the coil could be made to spin in the opposite direction. [1]

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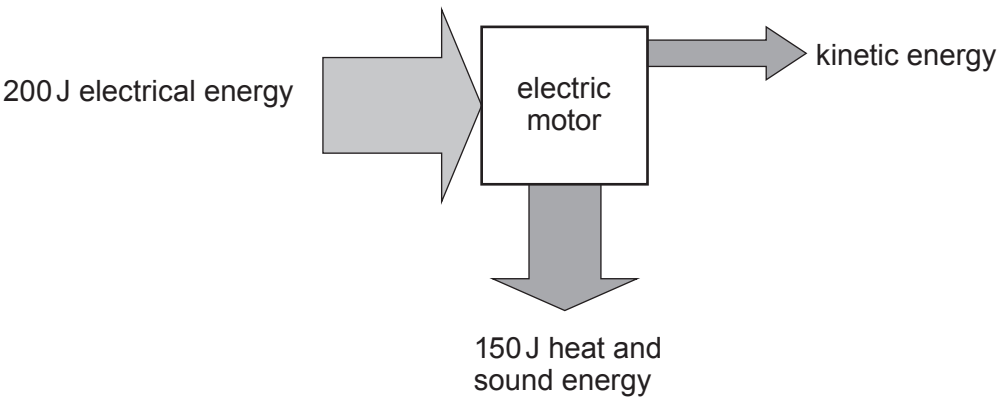
(ii) State **two** ways in which the coil could be made to spin faster. [2]

1.

2.



(c) A simple electric motor transfers energy as shown in the Sankey diagram below.



(i) Calculate how much useful energy the motor produces. [1]

Useful energy = J

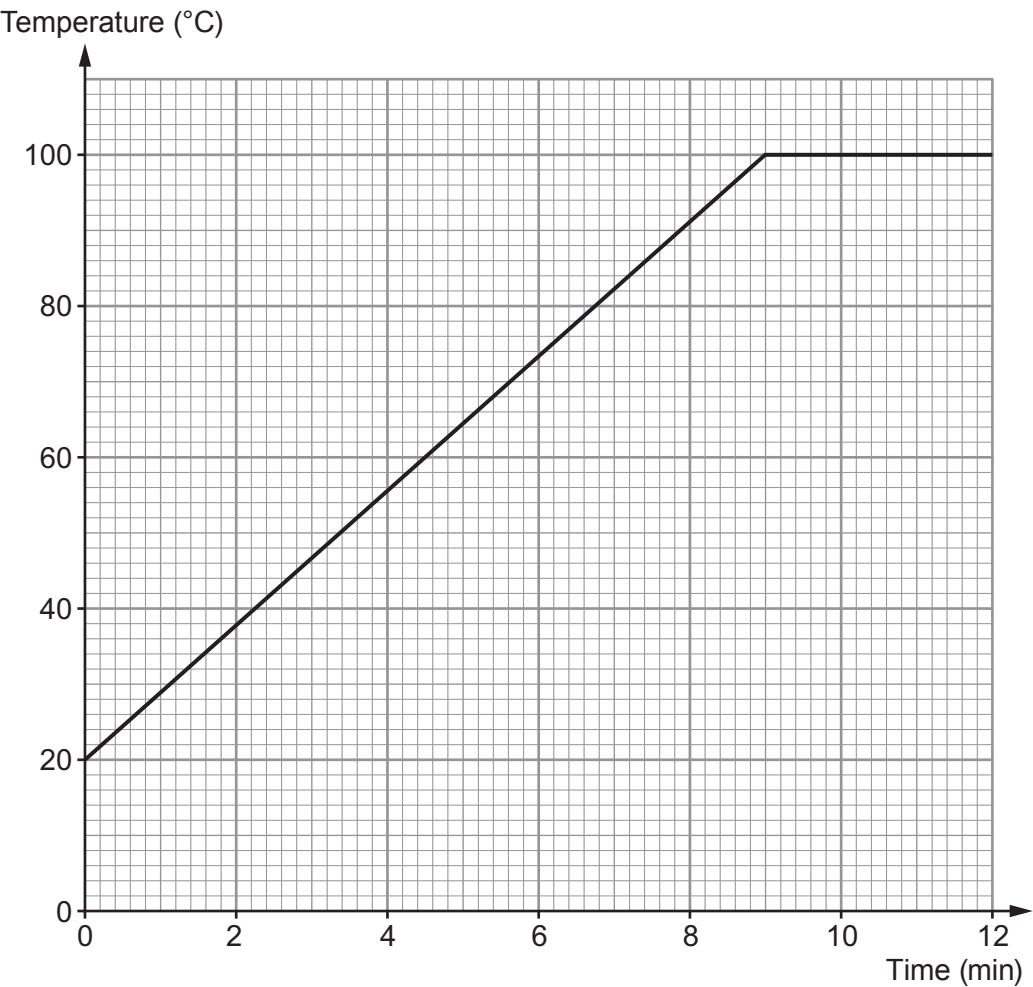
(ii) Use an equation from page 2 to calculate the % efficiency of the motor. [2]

% efficiency =

7



5. The graph below shows how the temperature of 0.2 kg of water changes as it is heated from 20 °C.



(a) Describe the relationship between temperature and time in the first 9 minutes. [2]

.....

.....

(b) (i) Write down the temperature of the water at 9 minutes. [1]

Temperature = °C

(ii) Calculate the change in temperature in the first 9 minutes. [1]

Change in temperature = °C



Examiner
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- (iii) The specific heat capacity of water is 4 200 J/kg °C.
Use the equation:

thermal energy = mass × specific heat capacity × change in temperature

to calculate how much thermal energy is supplied to the 0.2 kg water in the first 9 minutes. [2]

Thermal energy supplied = J

- (c) Between 9 and 12 minutes the water is boiling and its temperature stays constant even though heat energy is still being supplied. State what is happening to the water during this time. [1]

7

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6. Electricity in the UK is generated in a variety of ways. Most of our electricity is produced by burning fossil fuels, mainly gas and coal. When deciding which type of power station to build, it is important to consider the environmental problems they cause.

(a) Tick (✓) the **two** correct statements below. [2]

- Burning fossil fuels adds to climate change ☐
- Nuclear power stations emit lots of carbon dioxide when used ☐
- Tidal barrages damage marine habitats ☐
- Waste from gas power stations is radioactive ☐
- Wind power causes acid rain ☐

(b) The table below shows the gases released when the same mass of different fossil fuels are burned.

Fossil fuel	Emissions of polluting gas (units)		
	Carbon dioxide	Carbon monoxide	Sulfur dioxide
coal	208 000	208	2 591
oil	164 000	33	208
gas	11 700	40	1

Explain why coal has the greatest effect on global warming. [2]

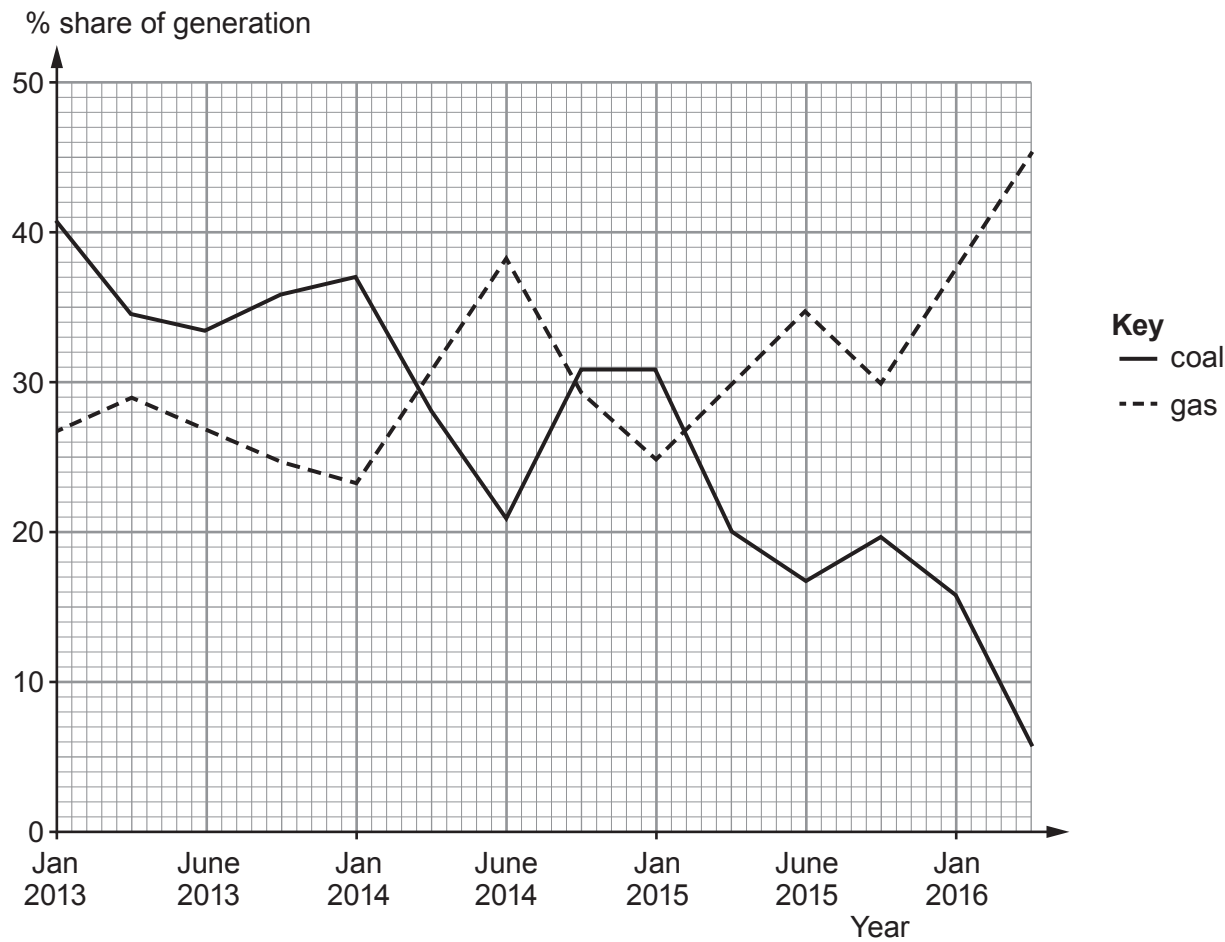
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(c) Coal and gas are both used in power stations to generate electricity. The diagram below shows how the percentage share of electricity generation from coal and gas has changed since 2013.



Use the graph to compare how the percentages of electricity generated from coal and gas changed during the time shown. [2]

.....

.....

.....



- (d) The UK is trying to increase the percentage of electricity generated by renewable sources such as wind.

Between April 2015 and April 2016 many new wind farms and other renewable power stations were built.
The table below shows the percentage of electricity generated by renewable sources in April 2015 and in April 2016.

Date	% of electricity produced by renewable sources	% of electricity produced by non-renewable sources
April 2015	25.4	74.6
April 2016	24.9	75.1

- (i) Use the data in the table to compare the percentage of electricity produced by renewable sources in April 2015 and April 2016. [1]

.....

- (ii) Is your answer to (d)(i) what you expected? Give a reason for your answer. [1]

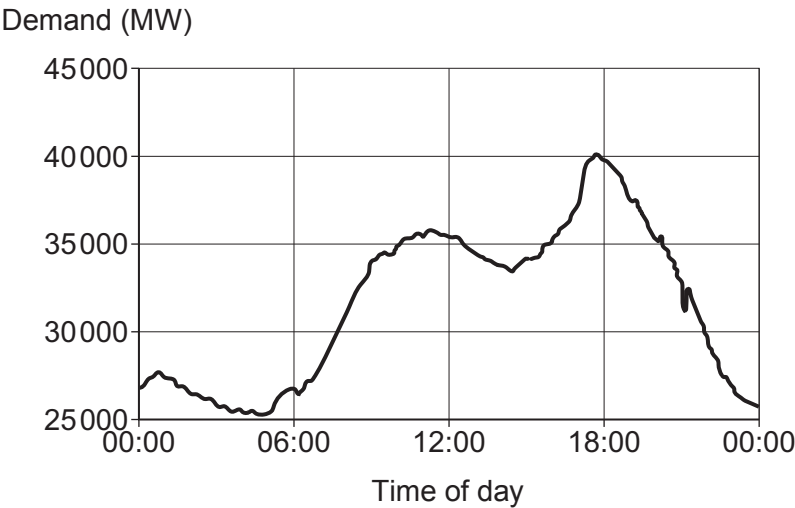
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Examiner
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(e) As well as generating electricity the UK also imports electricity from France. The diagram shows how demand for electricity varies throughout one day for the UK.



(i) Give a reason why electricity is imported into the UK. [1]

.....

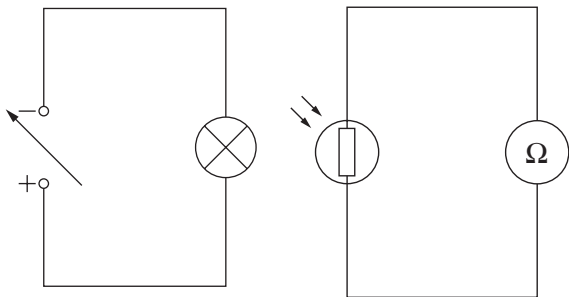
(ii) At what time of day is electricity most likely to be imported into the UK from France? [1]

.....

10



7. The following circuits are set up to investigate a light dependent resistor (LDR). The voltage of the power supply is changed to vary the power of the lamp to alter its brightness. The resistance of the LDR is measured with an ohmmeter (Ω) for each power of the lamp.



- (a) (i) State **two** variables, **other than using the same components**, that should be controlled in this experiment. [2]
1.
2.
- (ii) Explain how the design of the experiment could be improved to make the results more valid. [2]

.....

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.....

- (b) The results are shown in the table below.

Power of lamp (W)	Resistance of LDR (k Ω)
2	19.5
4	10.3
8	3.0
12	2.2
16	1.5
20	1.3
24	1.1



(i) Use the data to plot a graph on the grid below and draw a suitable line.

[4]

Examiner
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Resistance of LDR ($k\Omega$)



Examiner
only

(ii) Use the graph to find the resistance of the LDR for a lamp power of 10W. [1]

Resistance = Ω

(iii) It is suggested that when the lamp power doubles, the LDR resistance halves. Explain, using values from the table, to what extent this suggestion is true. [3]

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12



8. The **epicentre** is the point on the Earth's surface directly above an earthquake. Seismic stations detect earthquakes by the tracings made on seismographs.

- (a) Surface, P and S waves are three types of earthquake waves.
Tick (✓) the boxes next to the **three** correct statements about earthquake waves. [3]

Surface waves travel the fastest

☐

S waves travel on the surface of the Earth

☐

S waves are transverse waves

☐

P waves travel through solids and liquids

☐

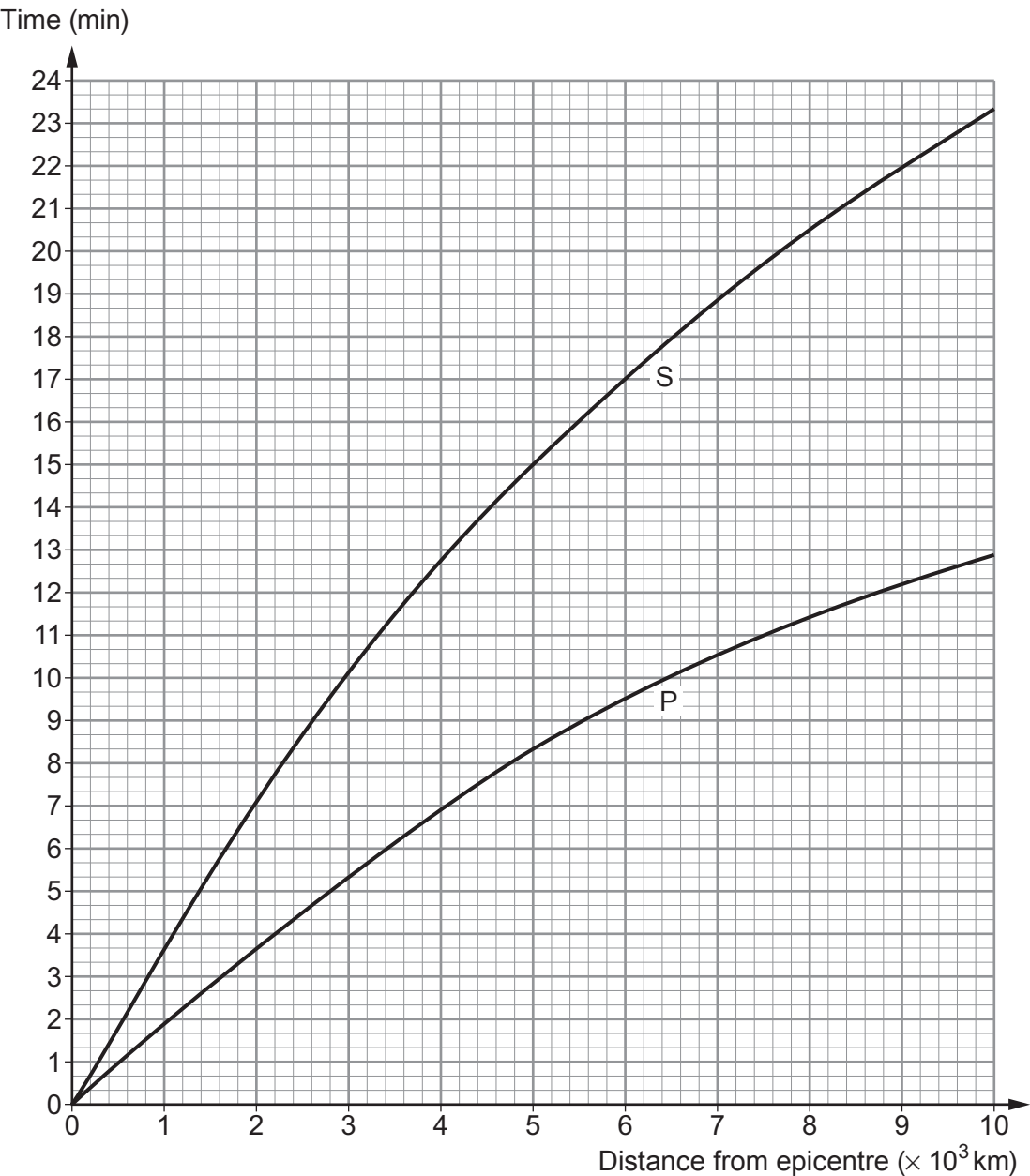
P waves are longitudinal waves

☐

S waves cause the most damage

☐

(b) The graph shows the time taken by P and S waves to travel different distances from the epicentre.



Each small square on the time axis represents 20 s.

(i) Use the graph to answer the following questions.

I. State the time it takes for a P wave to travel 5×10^3 km from the epicentre. [1]

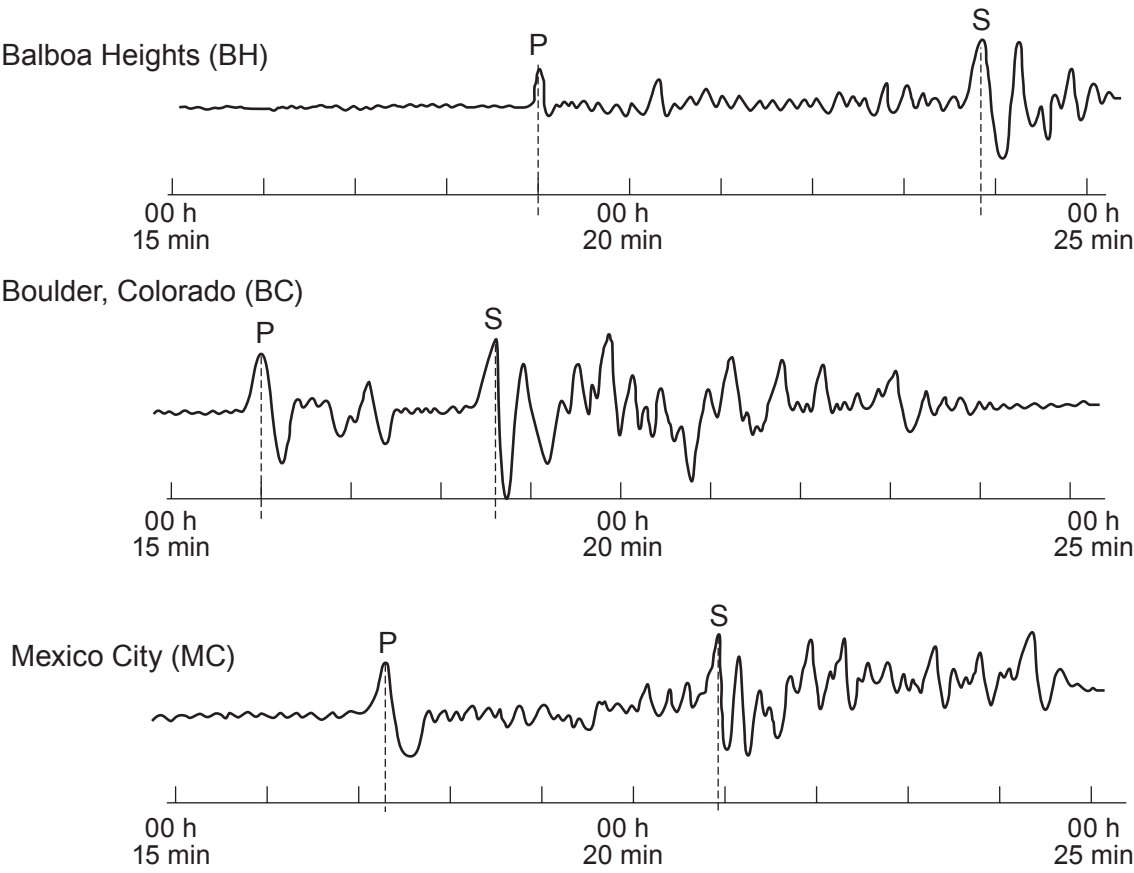
time = min s

II. State the **extra** time it takes S waves to travel 5×10^3 km from the epicentre. [1]

time = min s



(ii) Study the three seismograph tracings below. Tracings made at three separate seismic stations are needed to locate an earthquake epicentre. P shows the arrival of P waves and S shows the arrival of S waves.



Use the information in the graph and tracings to **complete the table**. [3]

City	Arrival time of P waves (h:min:s)	Arrival time of S waves (h:min:s)	Time difference for P and S waves (h:min:s)	Distance to epicentre ($\times 10^3$ km)
Balboa Heights (BH)	00:19:00	00:23:50	00:04:50	3.2
Boulder, Colorado (BC)	 : :	00:18:40	 : :	
Mexico City (MC)	00:17:15	00:20:55	00:03:40	2.2

TURN OVER FOR THE REST OF THE QUESTION



Examiner
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- (iii) The P waves arriving at Balboa Heights (BH) took 6 **minutes** to travel from the epicentre. Use an equation from page 2 to calculate the speed of the P waves arriving at Balboa Heights in **km/s**. [Note that $3.2 \times 10^3 \text{ km} = 3\,200 \text{ km}$] [2]

Speed = km/s

- (iv) The data is used to locate the epicentre of the earthquake. **Indicate with crosses (X) on the diagram opposite two** possible positions for the location of the earthquake. [1]
- (v) Use the data for Boulder Colorado (BC) to show clearly **on the diagram opposite** the actual location of the epicentre. Justify how you have arrived at your answer. [2]

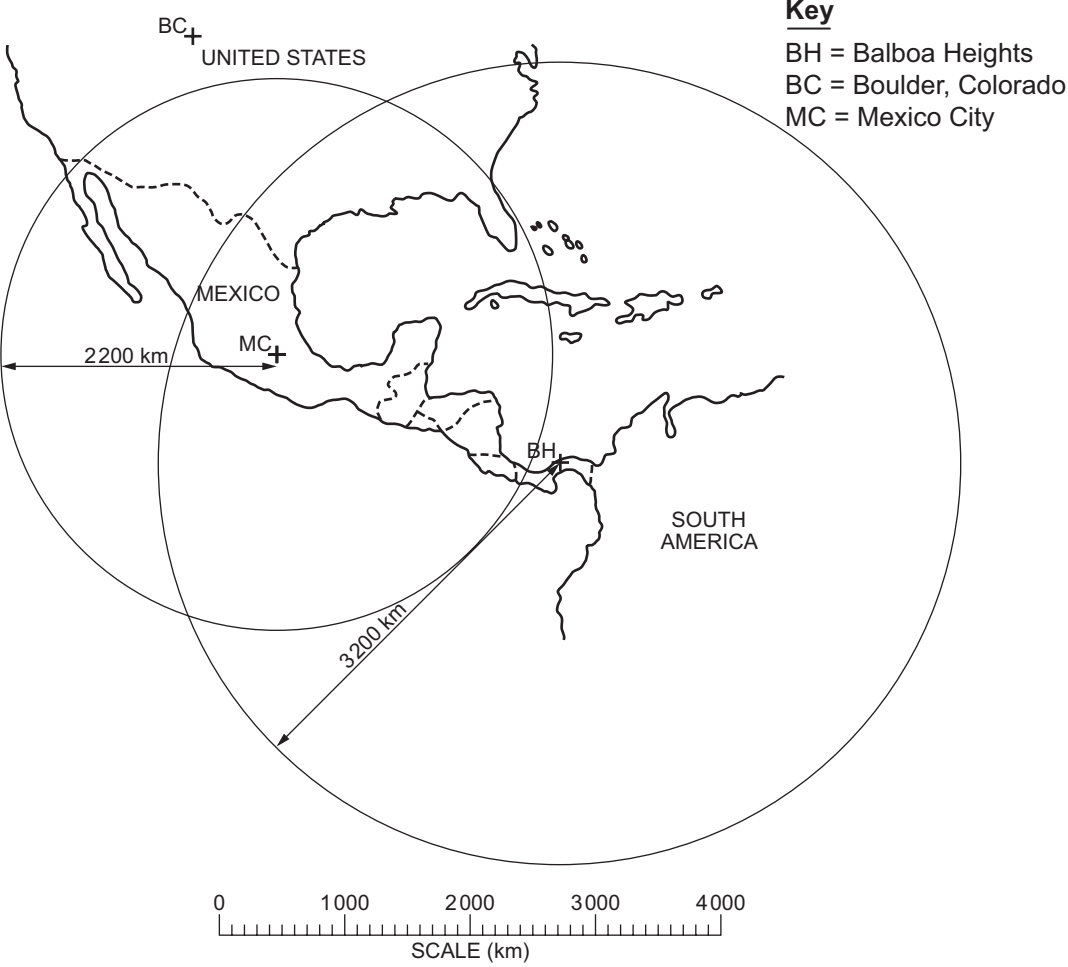
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13

END OF PAPER



Surname	Centre Number	Candidate Number
Other Names		0



GCSE

3420U10-1



FRIDAY, 14 JUNE 2019 – MORNING

PHYSICS – Unit 1:
Electricity, Energy and Waves

FOUNDATION TIER

1 hour 45 minutes

For Examiner’s use only		
Question	Maximum Mark	Mark Awarded
1.	5	
2.	8	
3.	15	
4.	16	
5.	6	
6.	10	
7.	12	
8.	8	
Total	80	

ADDITIONAL MATERIALS

In addition to this paper you will require a calculator and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen. Do not use correction fluid.
Write your name, centre number and candidate number in the spaces at the top of this page.
Answer **all** questions.
Write your answers in the spaces provided in this booklet. If you run out of space use the additional page at the back of the booklet.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.
The assessment of the quality of extended response (QER) will take place in question **3(a)**.



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Equations

current = $\frac{\text{voltage}}{\text{resistance}}$	$I = \frac{V}{R}$
total resistance in a series circuit	$R = R_1 + R_2$
energy transferred = power $\times$ time	$E = Pt$
power = voltage $\times$ current	$P = VI$
% efficiency = $\frac{\text{energy [or power] usefully transferred}}{\text{total energy [or power] supplied}} \times 100$	
density = $\frac{\text{mass}}{\text{volume}}$	$\rho = \frac{m}{V}$
units used (kWh) = power (kW) $\times$ time (h) cost = units used $\times$ cost per unit	
wave speed = wavelength $\times$ frequency	$v = \lambda f$
speed = $\frac{\text{distance}}{\text{time}}$	
pressure = $\frac{\text{force}}{\text{area}}$	$p = \frac{F}{A}$
change in thermal energy = mass $\times$ specific heat capacity $\times$ change in temperature	$\Delta Q = mc\Delta\theta$
thermal energy for a change of state = mass $\times$ specific latent heat	$Q = mL$
V_1 = voltage across the primary coil V_2 = voltage across the secondary coil N_1 = number of turns on the primary coil N_2 = number of turns on the secondary coil	$\frac{V_1}{V_2} = \frac{N_1}{N_2}$

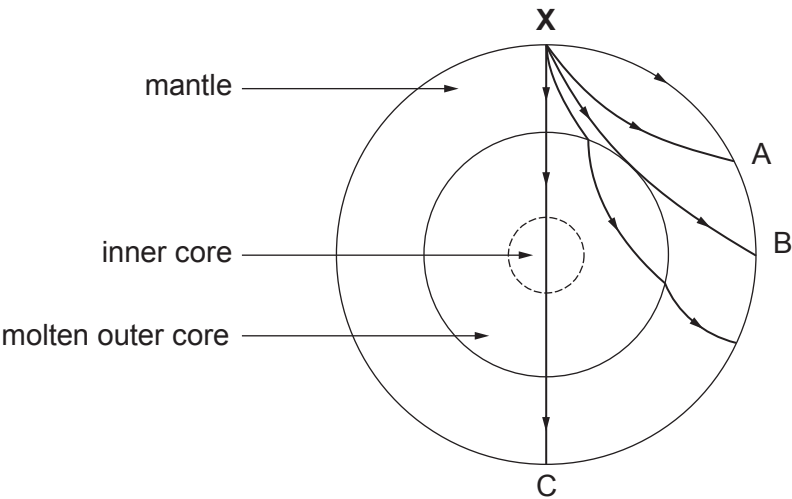
SI multipliers

Prefix	Multiplier
m	1×10^{-3}
k	1×10^3
M	1×10^6



Answer all questions.

1. Three types of wave spread from an earthquake. They are called P waves, S waves and surface waves. The diagram shows waves spreading out from an earthquake at X to monitoring stations A, B and C on the Earth's surface.



- (a) State **two** differences between P and S waves. [2]

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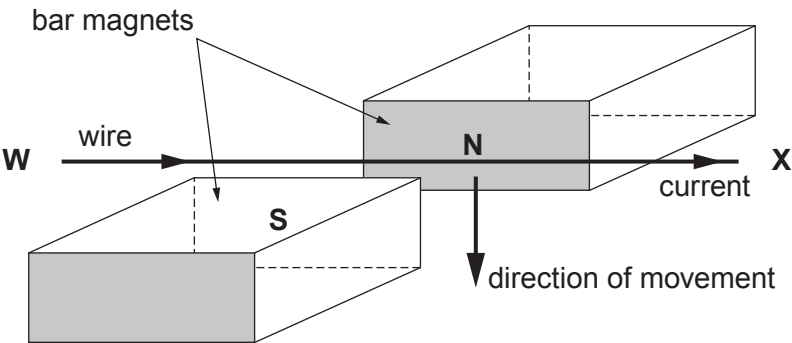
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- (b) Put a tick (✓) in the correct boxes in the following table to show which type of wave is detected at each point. The first row has been completed for you. [3]

Monitoring stations	P wave	S wave	Surface wave
A	✓	✓	✓
B			
C			



2. A teacher places a wire in a magnetic field. A current is passed through the wire and a force makes it move downwards. The following diagram shows how the apparatus was set up to show this effect.



A current is passed through the wire from **W** to **X**. A force makes the wire move **downwards**.

- (a) Tick (✓) the boxes alongside the **three** correct statements below. [3]

- The direction of the magnetic field is from N to S in the diagram.

☐
- The direction of movement is given by Fleming's right hand rule.

☐
- A current from **X** to **W** makes the wire move upwards.

☐
- When two South poles face each other the wire moves upwards.

☐
- A stronger magnetic field makes the wire move upwards.

☐
- This effect is used in electric motors.

☐

- (b) A student suggests to the teacher that the effect could be investigated further. Explain how the teacher could make changes that would affect the **size** of the force on the wire. [3]

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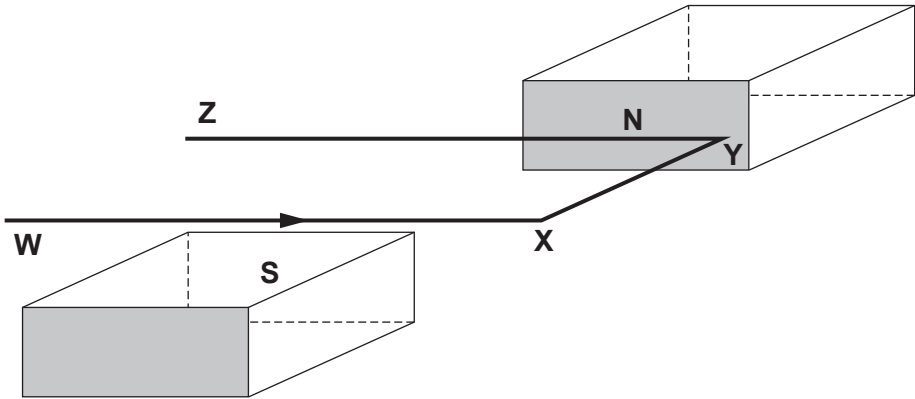
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- (c) In another experiment using the magnets, a single coil of wire **WXYZ** is placed between the magnets. A current is passed through the coil in the direction shown by the arrow.



Draw an arrow on the diagram to show the direction of the current in **YZ**.
Label the arrow C.

[1]

Draw a second arrow on the diagram to show the direction of the force on **YZ**.
Label the arrow F.

[1]

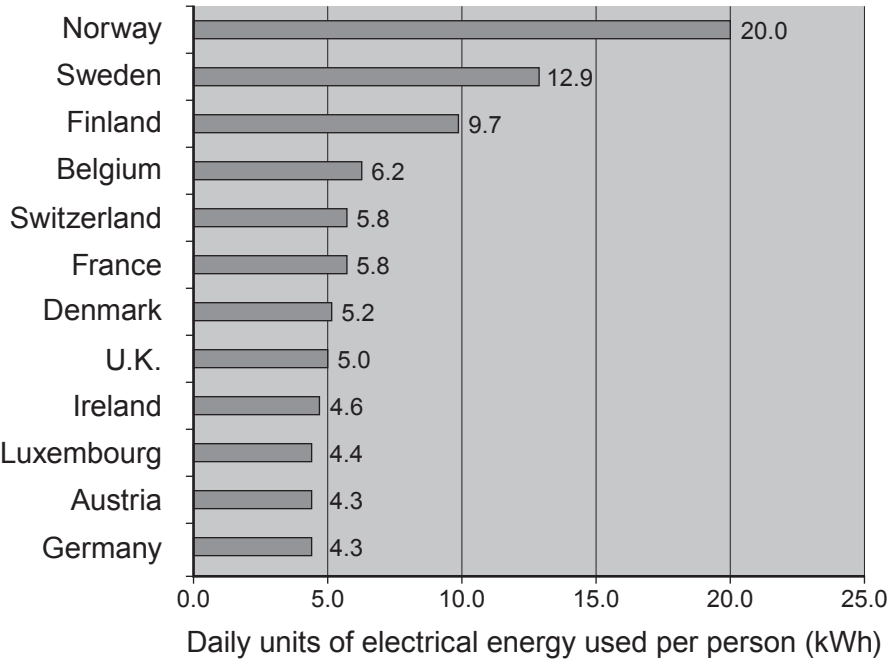
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05

8



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07

(b) The chart below shows the number of units of electricity used **each day per person** for the top twelve consuming nations in Europe for the year 2002.



Source “Energy Powering Your World” EFDA - European Fusion Development Agreement (2002)

(i) Name **one** country in which people use three times as much electrical energy as in Germany. [1]

.....

(ii) Norway is closer than France to the Earth’s North pole. Explain how this may account for the different amounts of electricity consumed in these countries. [2]

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(iii) The cost of a unit of electricity in Norway is half of the U.K. cost. Ann suggests that a person’s electricity bill in Norway would be half of that paid in the U.K. Use information in the chart to explain whether you agree with this suggestion. [2]

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Examiner
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(iv) In 2002, the average family in the U.K. consisted of three people.

I. Calculate the number of units used by this family in **one** week. [2]

Units used = kWh

II. Use an equation from page 2 to calculate how much this family would pay for electricity in a week in the U.K. in 2002.
The cost of one unit of electricity = 8 p. [2]

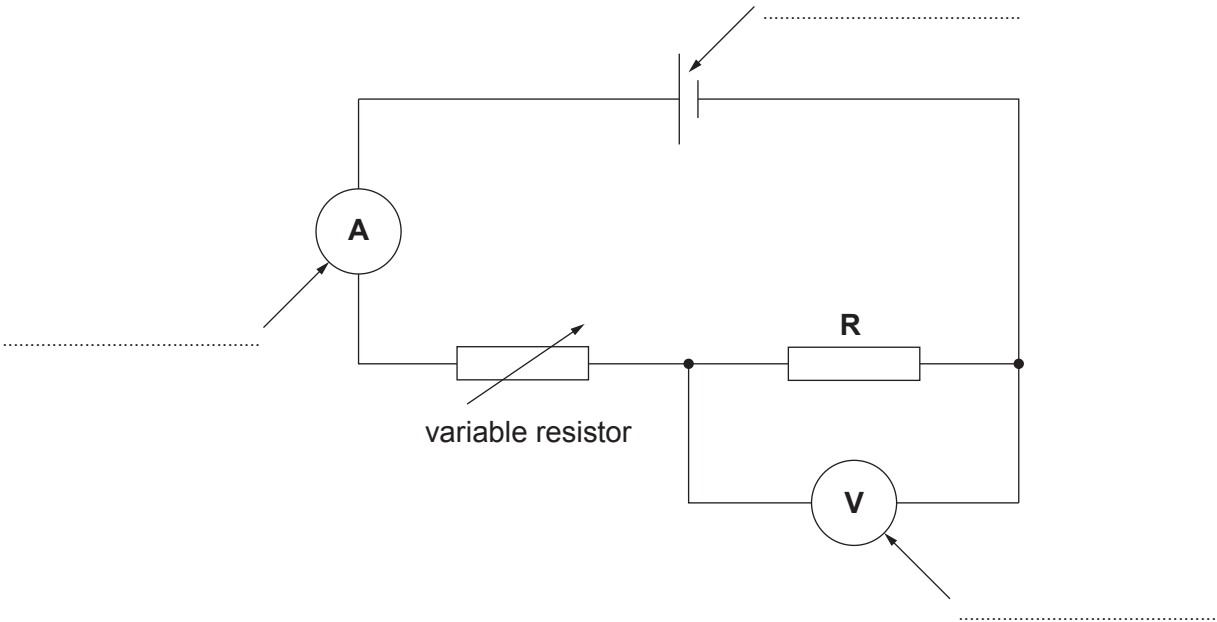
Cost = p

15

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09



4. Some pupils set up the circuit below to measure the current in a resistor (**R**) and the voltage across it. The circuit is used to give a number of readings.



- (a) (i) **Label** the parts of the circuit diagram shown above. [3]
(ii) State the purpose of the variable resistor. [1]

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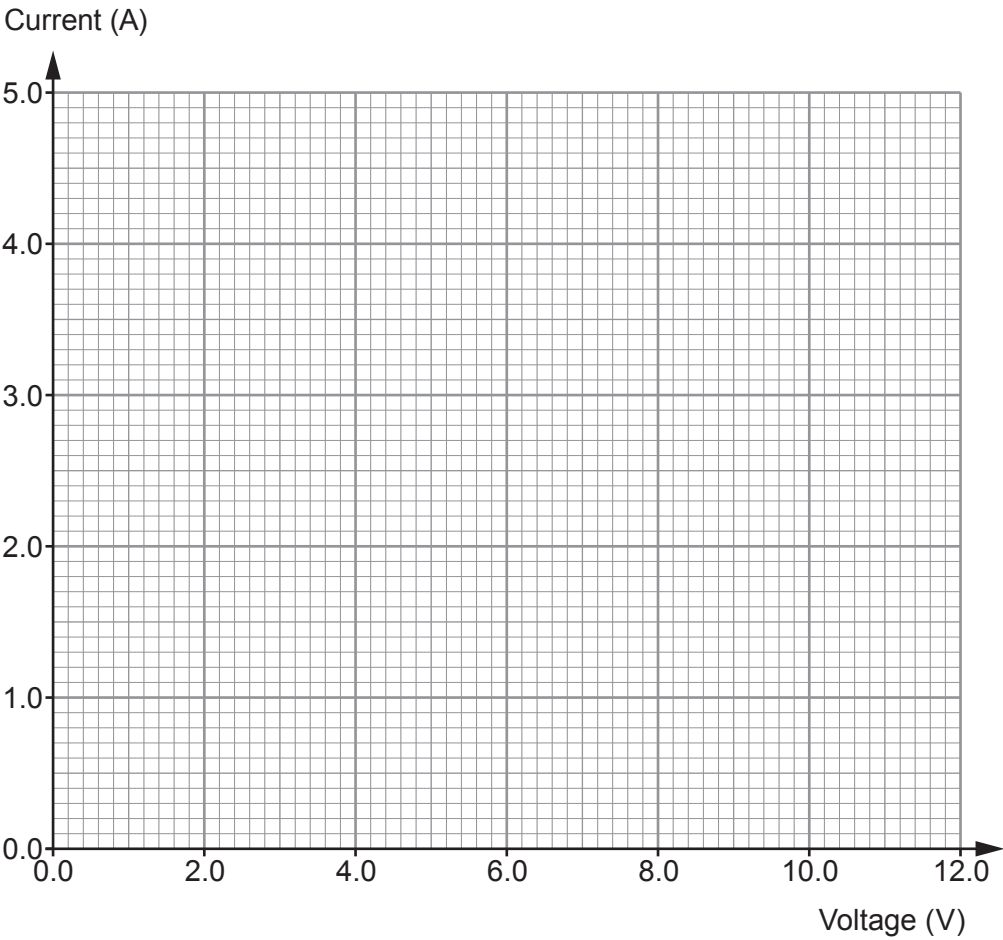


(b) The circuit was used to obtain the following results.

Voltage (V)	Current (A)
0.0	0.0
2.0	0.8
4.0	1.6
6.0	2.4
10.0	4.0
12.0	4.8

(i) Plot the data on the grid below and draw a suitable line.

[3]



(ii) Use your graph and the equation:

$$\text{resistance} = \frac{\text{voltage}}{\text{current}}$$

to calculate the resistance of the resistor at 8.0V. [2]

Resistance = Ω

(iii) Complete the following sentence about the trend on the graph by underlining the correct word or phrase in each bracket. [3]

As the voltage increases, the current (**decreases** / **stays the same** / **increases**)
at a (**decreasing** / **constant** / **increasing**) rate. As a result, the resistance
(**decreases** / **stays constant** / **increases**).

(iv) One pupil in the group, Billy, made the statement that the power of the resistor increases as the voltage increases. By using an equation from page 2, explain whether you agree with Billy. [2]

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(v) A second group of pupils use a resistor that has **half** the resistance of the one in this experiment. **Draw a line on the original grid** to show how the current through it would change with voltage. [2]

16



5. Satellites can be positioned in orbit to communicate with base stations on the ground. Messages can be relayed around the world in this way. The diagram shows base stations **A**, **B**, **C** and **D** and four satellites.

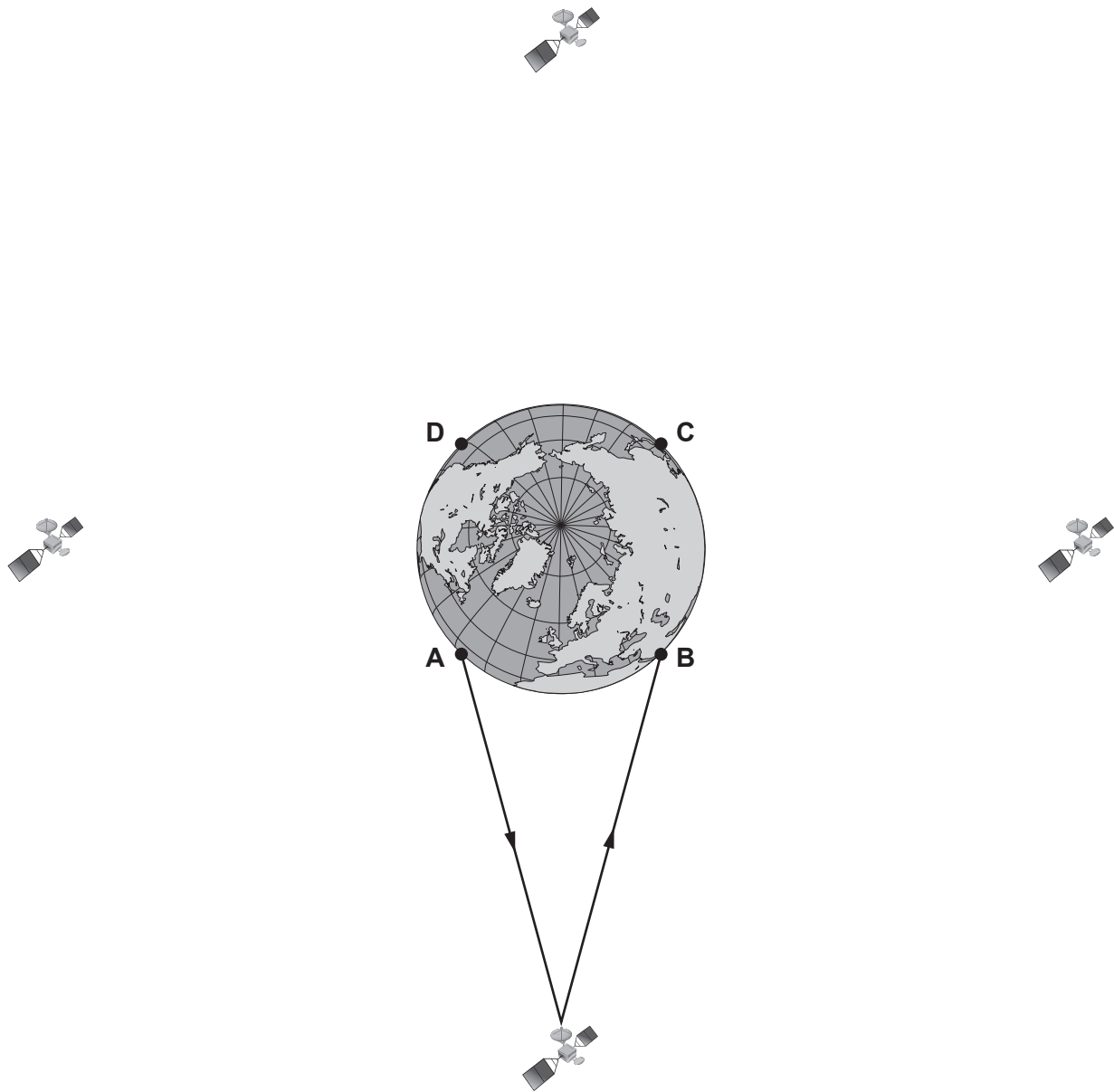


Diagram is not drawn to scale

- (a) The diagram shows the path of a signal sent from base station **A** to base station **B**. **Continue the path of the signal** so that it can be received at base station **C**. [1]



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- (b) The distance from a base station to a satellite is 36 000 km. Calculate the total distance travelled by the signal in going from **A** to **C**.
Give your answer in **metres**. [3]

Total distance = m

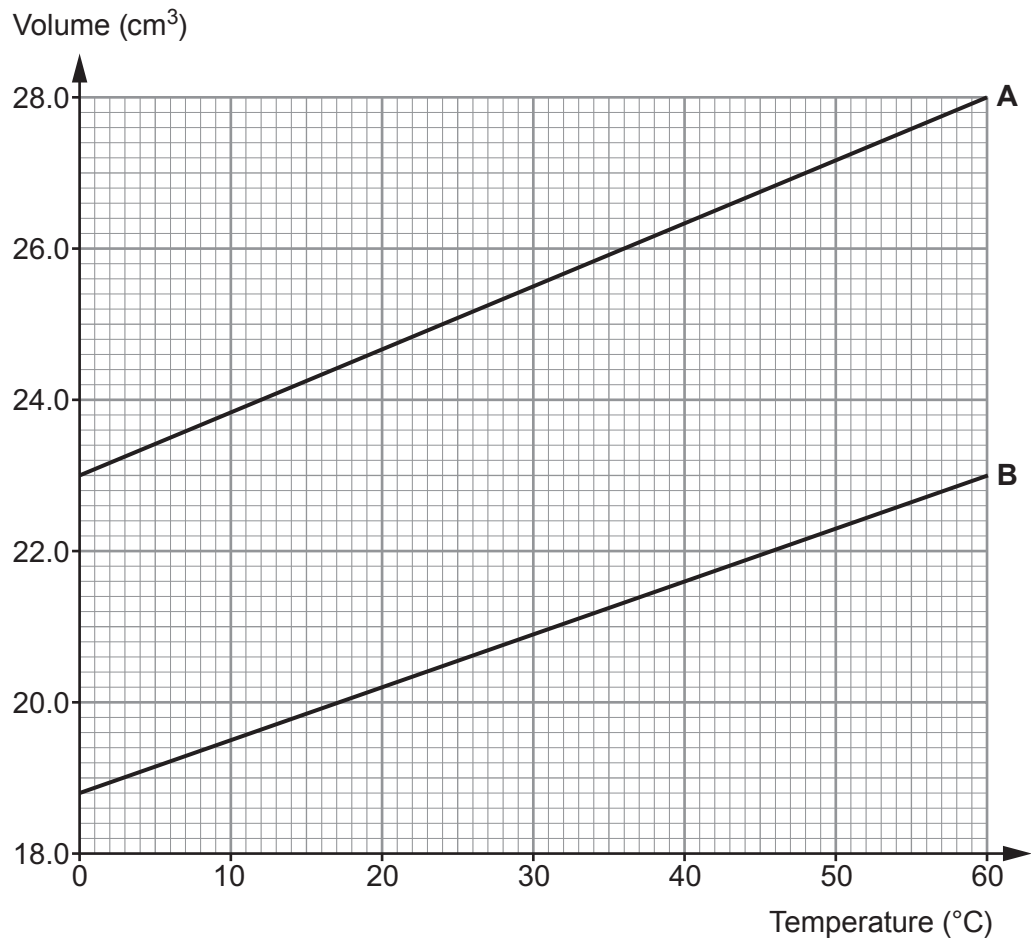
- (c) The time taken for the signal to travel from **A** to **C** is 0.48 s. Use an equation from page 2 to calculate the speed of the signal. [2]

Speed = m/s

6



6. In a class experiment, the volumes of two different masses of trapped air (**A** and **B**) were measured as their temperature was increased from 0 °C to 60 °C. The results are shown in the graphs below.



- (a) (i) At what temperature is the volume of gas **B** equal to 22.0 cm³? [1]
Temperature = °C
- (ii) The difference in the volumes of the two gases at 0 °C is 4.2 cm³. Use the graph to calculate the difference in their volumes at 60 °C. [1]
Difference in volumes = cm³
- (iii) Both graph lines are straight. Using your answer to (ii) above, compare their gradients. [1]

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Examiner
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- (b) (i) If both lines are extended to lower temperatures, they would meet at -273°C . This temperature is zero on the Kelvin scale of temperature. State the name given to this temperature. [1]

.....

- (ii) Calculate the highest temperature **on the Kelvin scale** reached by the gases in this experiment. [1]

Temperature = K

- (c) In another experiment, gas **B** is kept in a container of fixed volume. State what happens, if anything, to the pressure of the gas as its temperature is increased. [1]

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- (d) (i) A gas exerts a force of $180\,000\text{ N}$ on the walls of its container. The surface area of the container is 1.5 m^2 . Use an equation from page 2 to calculate the pressure of the gas. [2]

Pressure = Pa

- (ii) The gas is kept in a container at constant pressure. Julia says that if the area is doubled, the force would halve. Explain whether Julia's statement is correct. [2]

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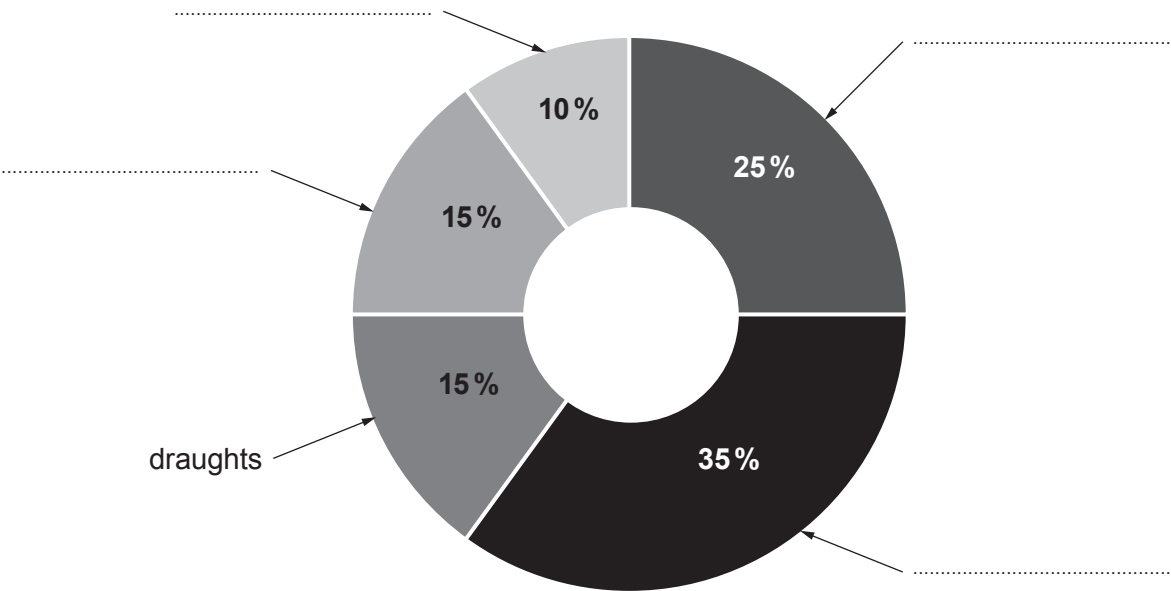


7. The Welsh Government Warm Homes Scheme, called Nest, aims to make Welsh homes warmer and more energy efficient places to live. Nest is accessible to all homeowners in Wales and provides advice on saving energy.

(a) Draughts, floors, windows, walls and the roof are the five ways energy is lost from a heated house. The pie chart shows the percentages of energy loss by each of the five ways.

Use the information below to **complete the labelling** on the pie chart. [3]
One label has been completed already.

- The percentage energy loss through the walls is greatest.
- Windows and draughts have equal percentage energy losses.
- Windows and floor percentage losses add up to equal the percentage loss through the roof.



(b) To reduce energy loss through the roof the Nest Scheme suggests the installation of fibre-glass insulation in the loft.

(i) Explain how fibre-glass reduces energy loss in the loft by convection. [2]

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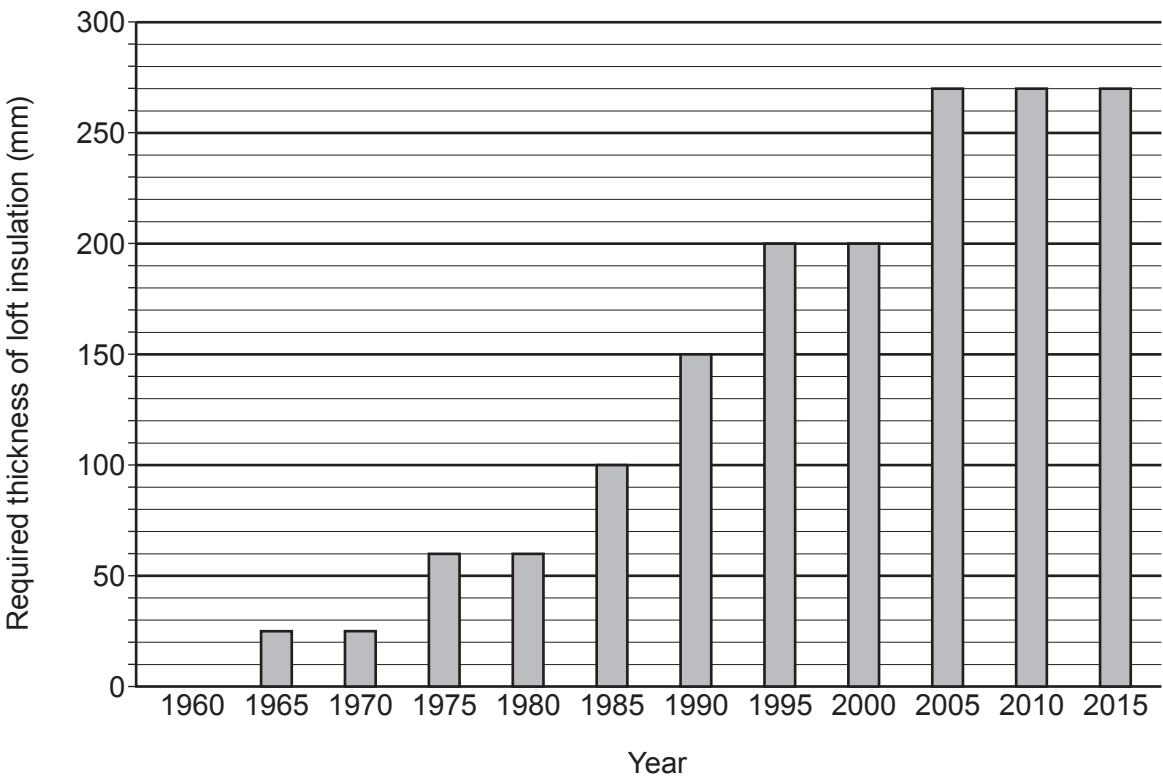
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- (ii) There are strict regulations about the minimum thickness of insulation that must be installed in the loft of a new house. The graph shows how the required thickness of loft insulation has changed since 1960.



Use information from the graph to tick (✓) the **three** correct statements. [3]

- Every five years the required thickness of loft insulation increases. ☐
- The required thickness of loft insulation in 2000 is 8 times thicker than in 1970. ☐
- In 1960 houses did not lose any energy through their roof. ☐
- A house built in 1980 needs 210 mm of loft insulation added to bring it up to 2015 standards. ☐
- The **general** trend of the graph indicates that the required thickness of loft insulation has increased at an increasing rate. ☐
- The required thickness of loft insulation will remain constant after 2015. ☐



Examiner
only

- (iii) A homeowner must install loft insulation in a new extension. **It has a loft area of 120 m².** The insulation must be at least 270 mm thick to meet building regulations.

There is a selection of fibre-glass insulations available.

	Insulation 1 (270 mm thick)	Insulation 2 (350 mm thick)	Insulation 3 (300 mm thick)
Installation cost (£/m ²)	3.50	5.55	4.50
Estimated saving per year (£)	84	111	98
Payback time (years)	5.0	6.0	

Calculate the payback time if **insulation 3** was installed in the 120 m² extension. [2]

Payback time = years

- (iv) The homeowner considers installing **insulation 1** as it is cheapest but the builder says that **insulation 2** should be installed as it will save more money over 40 years. Explain, with calculations, whether the builder is correct. [2]

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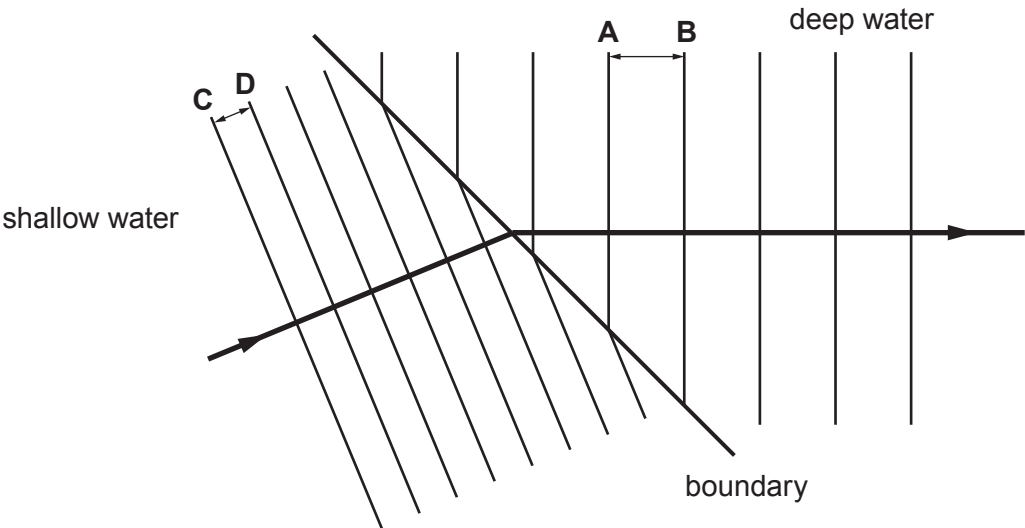
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8. In class, a teacher demonstrates refraction using a ripple tank. The diagram below shows plane wavefronts travelling across a boundary between shallow and deep water. The frequency of the waves remains **constant** during refraction.



- (a) Using a ruler, students measure the distance between wavefronts **A** and **B**. This measurement is the wavelength of the water waves in deep water. The distance between wavefronts **C** and **D** is measured to obtain their wavelength in the shallow water. The results are shown below.

	Deep water (AB)	Shallow water (CD)
Wavelength (mm)	10	5

- (i) State how the measurement of wavelength could be improved. [1]

.....

.....

- (ii) The wavelength in the deep water is twice the wavelength in the shallow water. The teacher suggests, “the speed of the wavefronts in shallow water is double the speed of the wavefronts in the deep water.” Using information provided explain if the suggestion made by the teacher is correct. [2]

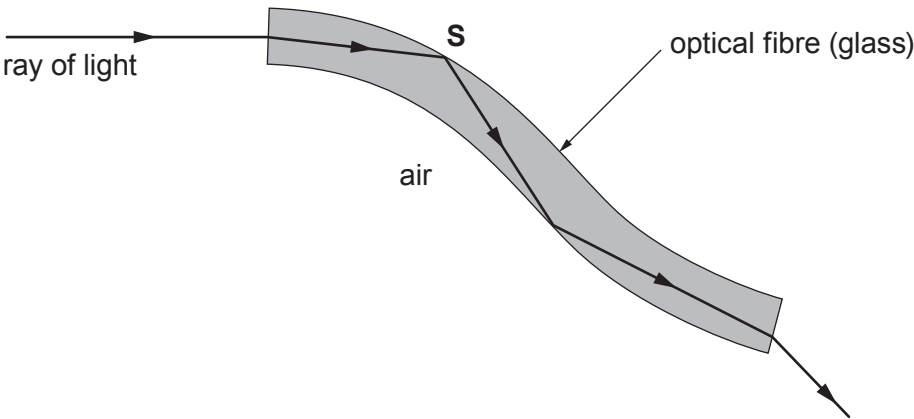
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- (b) An endoscope uses optical fibres. It can be used by doctors to produce medical images of a specific area inside a patient. A bundle of fibres is inserted into the body. Some of the fibres carry light into the body and others return the light reflected off internal surfaces. The diagram shows a ray of light passing through part of an optical fibre of an endoscope.



- (i) State the name given to the change in direction of the signal at **S**. [1]

.....

- (ii) State the **two** conditions needed for the ray of light to change direction at **S**. [2]

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- (iii) Medical images can also be obtained from a computer tomography (CT) scan. This type of scan uses X-rays targeted at the patient from different positions outside the body. The information collected is processed by a computer to produce detailed 3D image segments of the patient.
Explain a disadvantage of using a CT scan to obtain medical information compared to using an endoscope. [2]

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END OF PAPER



Surname	Centre Number	Candidate Number
First name(s)		0

**GCSE**

3420U10-1



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MONDAY, 20 JUNE 2022 – MORNING

PHYSICS – Unit 1:
Electricity, Energy and Waves

FOUNDATION TIER

1 hour 45 minutes

For Examiner's use only		
Question	Maximum Mark	Mark Awarded
1.	10	
2.	12	
3.	6	
4.	13	
5.	12	
6.	7	
7.	7	
8.	13	
Total	80	

3420U101
01**ADDITIONAL MATERIALS**

In addition to this paper you will require a calculator and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen or correction fluid.

You may use a pencil for graphs and diagrams only.

Write your name, centre number and candidate number in the spaces at the top of this page.

Answer **all** questions.

Write your answers in the spaces provided in this booklet. If you run out of space use the additional page at the back of the booklet, taking care to number the question(s) correctly.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.

The assessment of the quality of extended response (QER) will take place in question **5(a)**.



JUN223420U10101

Equations

current = $\frac{\text{voltage}}{\text{resistance}}$	$I = \frac{V}{R}$
total resistance in a series circuit	$R = R_1 + R_2$
energy transferred = power $\times$ time	$E = Pt$
power = voltage $\times$ current	$P = VI$
% efficiency = $\frac{\text{energy [or power] usefully transferred}}{\text{total energy [or power] supplied}} \times 100$	
density = $\frac{\text{mass}}{\text{volume}}$	$\rho = \frac{m}{V}$
units used (kWh) = power (kW) $\times$ time (h) cost = units used $\times$ cost per unit	
wave speed = wavelength $\times$ frequency	$v = \lambda f$
speed = $\frac{\text{distance}}{\text{time}}$	
pressure = $\frac{\text{force}}{\text{area}}$	$p = \frac{F}{A}$
change in thermal energy = mass $\times$ specific heat capacity $\times$ change in temperature	$\Delta Q = mc\Delta\theta$
thermal energy for a change of state = mass $\times$ specific latent heat	$Q = mL$
V_1 = voltage across the primary coil V_2 = voltage across the secondary coil N_1 = number of turns on the primary coil N_2 = number of turns on the secondary coil	$\frac{V_1}{V_2} = \frac{N_1}{N_2}$

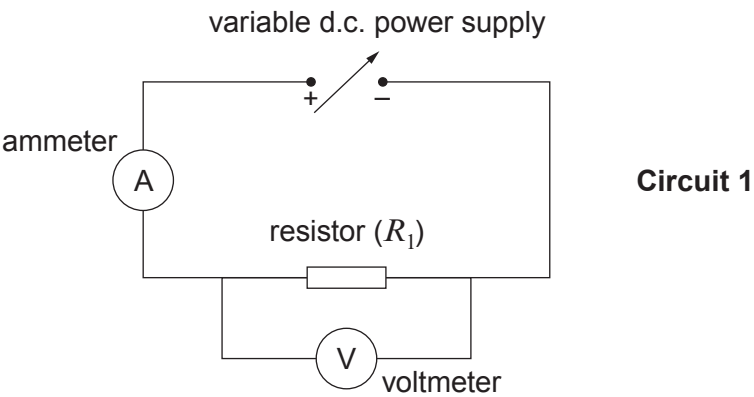
SI multipliers

Prefix	Multiplier
m	1×10^{-3}
k	1×10^3
M	1×10^6



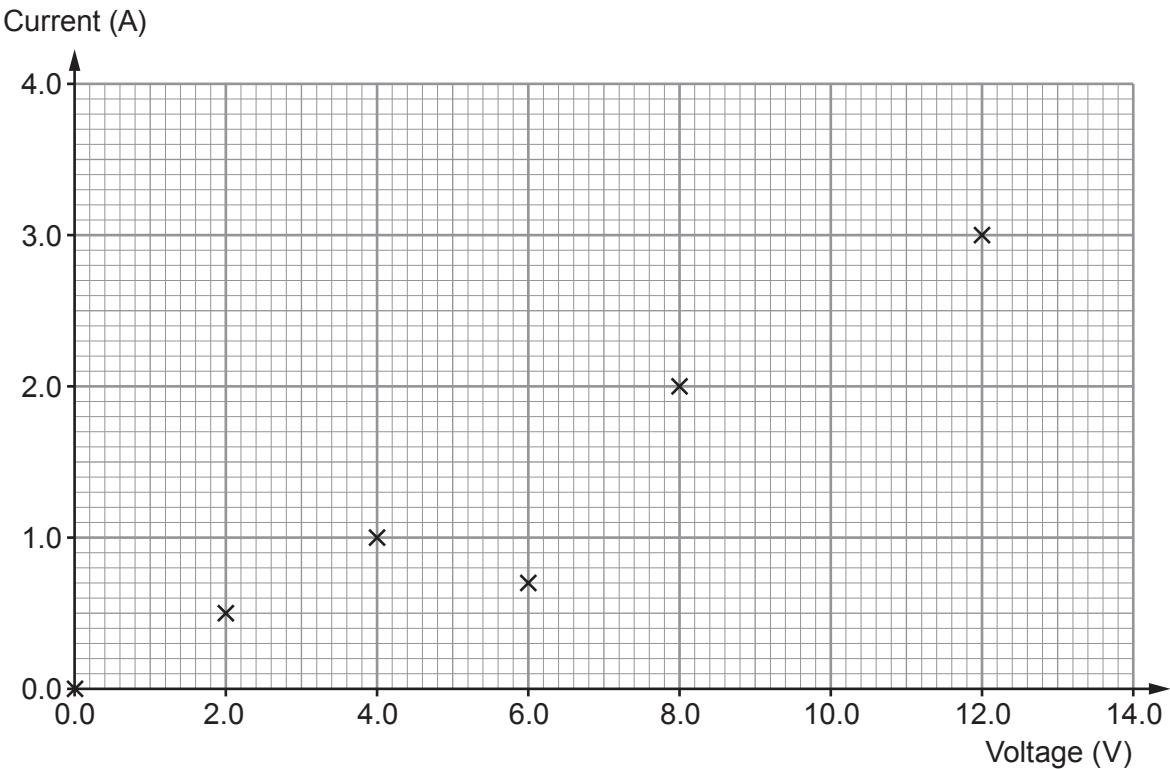
Answer **all** questions.

1. Paul investigated how the current changes with voltage for a resistor (R_1). The circuit he used is shown (**Circuit 1**).



(a) Name the device he used to measure the current. [1]

(b) Paul's results are shown on the graph.



(i) On the graph, **circle** the anomalous point. [1]

(ii) **Complete the graph** by drawing a suitable straight line. [1]



(iii) I. Use the graph to state the current at 12.0 V. [1]

Current = A

II. Use the equation:

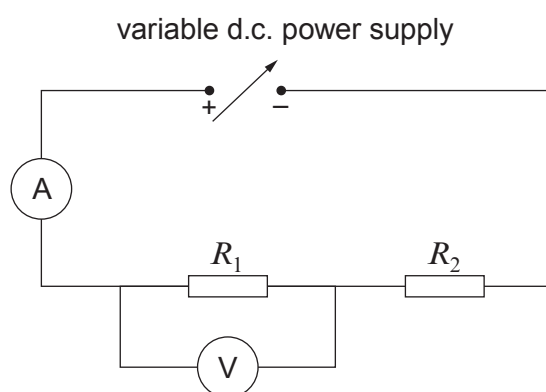
$$\text{resistance} = \frac{\text{voltage}}{\text{current}}$$

to calculate the resistance of the resistor (R_1) at 12.0 V. [2]

Resistance of R_1 = Ω

(iv) State what the graph shows about the resistance of the resistor (R_1) as the voltage changes. [1]

- (c) The original circuit is then adapted by adding a second, **identical** resistor (R_2) in series with R_1 .
This circuit is shown below (**Circuit 2**).



Circuit 2

Complete the following sentences by underlining the correct word or phrase in each bracket. [3]

The total resistance in circuit 2 is (**double / the same as / half**) the total resistance of circuit 1.

When the d.c. power supply is set at the same voltage, the voltmeter reading in circuit 2 is (**double / the same as / half**) the voltmeter reading in circuit 1.

The current in circuit 2 is (**double / the same as / half**) the current in circuit 1.



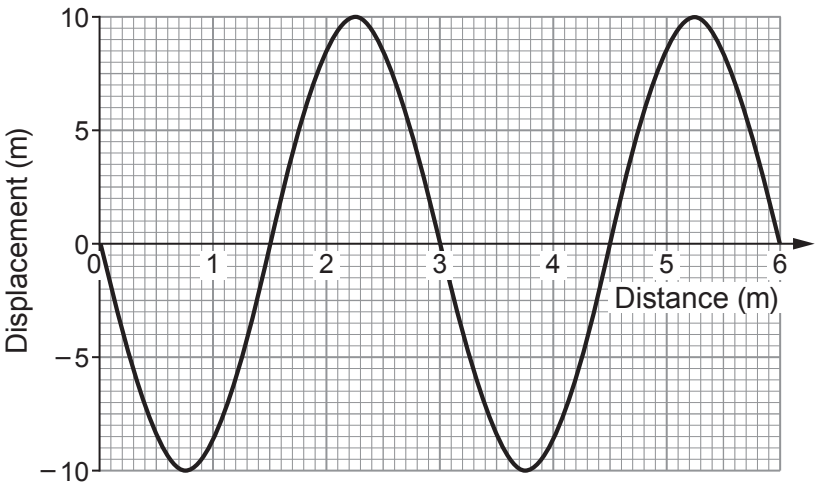
2. The diagram shows regions of the electromagnetic (em) spectrum.
The regions are in order.

radio waves	microwaves	infra-red	visible light	ultraviolet	X-rays	gamma rays
-------------	------------	-----------	---------------	-------------	--------	------------

(a) **Tick (✓) the *three* correct statements.** [3]

- The frequency of em waves decreases from radio waves to gamma rays. ☐
- All em waves travel at the same speed in a vacuum. ☐
- Gamma rays can be emitted from radioactive materials. ☐
- The wavelength of em waves decreases from radio waves to gamma rays. ☐
- Radio waves, microwaves, infra-red and visible light are all ionising radiations. ☐
- All em waves are longitudinal. ☐

(b) The graph below shows a wave diagram.



Use the following values to answer the questions below. You may use each value once, more than once or not at all.

2	3	5	6	10
---	---	---	---	----

- (i) I. State the amplitude of the wave. m [1]
- II. State the number of complete waves shown. waves [1]
- III. State the wavelength of the wave. m [1]



(ii) I. Complete the following sentences by underlining the correct phrase in each bracket. [2]

The wave has a frequency of (7.5Hz / 7.5cm/s / 7.5cm).

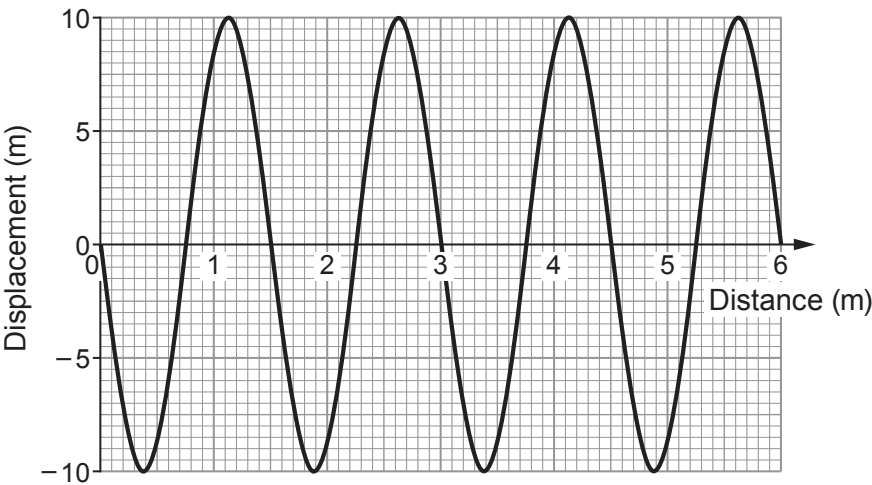
This means 7.5 waves pass a point each (second / minute / hour).

II. Use information from parts (b)(i) and (ii) to calculate the wave speed.
Use the equation:

wave speed = wavelength $\times$ frequency [2]

Wave speed = m/s

(c) The speed of the wave remains constant but its **frequency is doubled**.
A student draws a diagram of the expected wave. It is shown below.



Explain whether you agree with the student's diagram of the wave. [2]

.....

.....

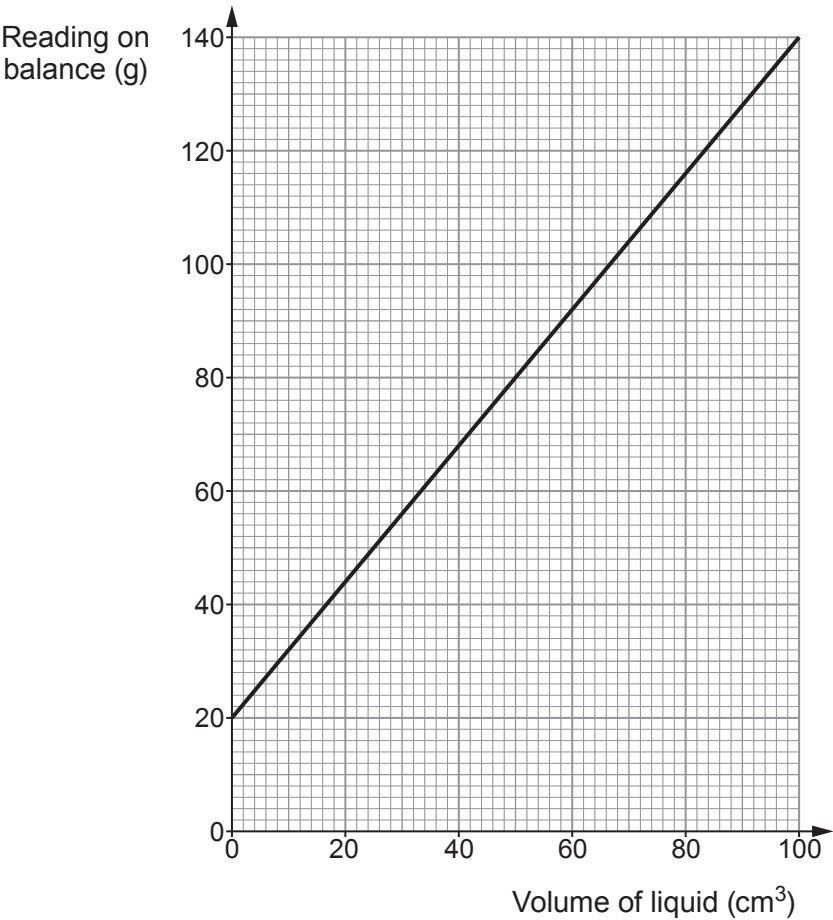
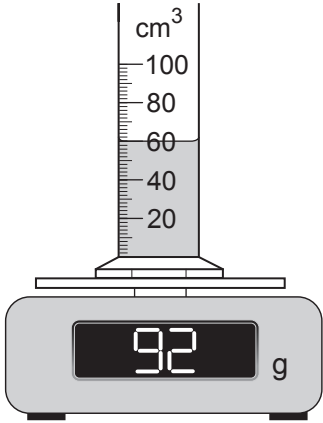
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3. Kylie carries out an experiment, in class, to measure the density of a liquid. She places an empty measuring cylinder on a balance and pours the liquid into it. She records the reading from the balance for different volumes of liquid. Her results are shown in the table and on the graph.

Volume of liquid (cm ³)	Reading on balance (g)
20	44
40	68
60	92
80	116
100	140



Use the information above to answer the following questions.

- (a) State the mass of the empty measuring cylinder. g

[1]



Examiner
only

(b) Calculate the mass of liquid in the measuring cylinder when its volume is 100 cm³. [2]

Mass of liquid = g

(c) Use an equation from page 2 to calculate the density of the liquid that Kylie used and state the unit of your answer. [3]

Density =

Unit =

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6



4. A Welsh farmer decides to purchase a wind generator to provide some electrical energy for his farm.
A selection of wind generators is available.

	Wind generator 1	Wind generator 2	Wind generator 3
Cost to buy and install (£)	1 800	3 000	2 925
Expected lifetime (years)	15	15	15
Estimated number of units produced per year (kWh)	2 000	4 000	3 000
Cost per unit (£)	0.15	0.15	0.15
Estimated saving per year (£)		600	450
Estimated payback time (years)		5.0	6.5

- (a) (i) Use the equation:

$$\text{estimated saving per year} = \text{units produced per year} \times \text{cost per unit}$$

and data from the table to calculate the estimated saving per year if **wind generator 1** was installed.

[1]

Estimated saving per year = £

- (ii) Use the equation:

$$\text{estimated payback time} = \frac{\text{cost to buy and install}}{\text{estimated saving per year}}$$

and data from the table to calculate the estimated payback time for **wind generator 1**.

[2]

Estimated payback time = years



- (b) (i) A salesperson suggests to the farmer that **wind generator 3** is better than **wind generator 2** because it is cheaper.
Explain, using information from the table, if the salesperson's suggestion is correct over a 5-year timescale. [2]

.....

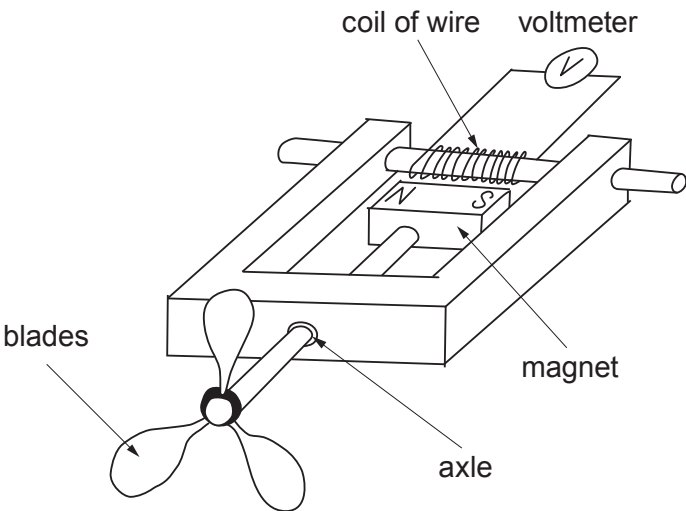
.....

.....

- (ii) Give **two** reasons why the payback time can only be an estimate. [2]

1.
2.

- (c) A wind generator only operates in certain wind speeds.
The diagram below shows a student's design for measuring wind speed.



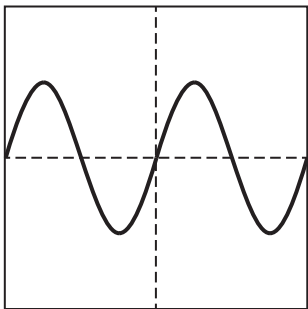
- (i) When the wind blows a reading on the voltmeter is obtained.
The following sentences describe the different stages.
- A** The magnetic field lines cut through the coil of wire.
 - B** The blades rotate.
 - C** The magnet rotates.
 - D** This induces a voltage in the coil of wire.

Put the letters **A**, **B** and **C** in the boxes to correctly describe the process of inducing a voltage in the coil of wire. The final box has already been completed, [2]

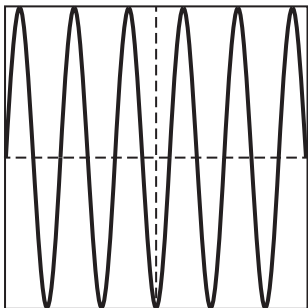
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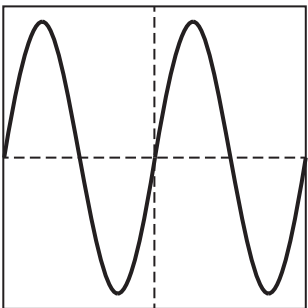
(ii) When the wind speed is **2 m/s** the alternating voltage produced is shown below.



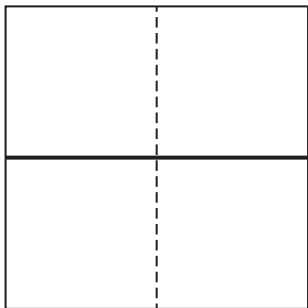
The following traces **X**, **Y** and **Z** are produced during different wind conditions.



Trace **X**



Trace **Y**



Trace **Z**

I. State which trace (**X**, **Y** or **Z**) is produced when the wind speed is **faster than 2 m/s**.

Trace [1]

II. State which trace (**X**, **Y** or **Z**) is produced when **the wind stops**.

Trace [1]

(d) When the student's design is built it is found that the voltmeter is not sensitive enough to measure a wind speed less than 0.5 m/s.

(i) Suggest **one** change that can be made to the **magnet** so that wind speeds less than 0.5 m/s can be measured by the voltmeter. [1]

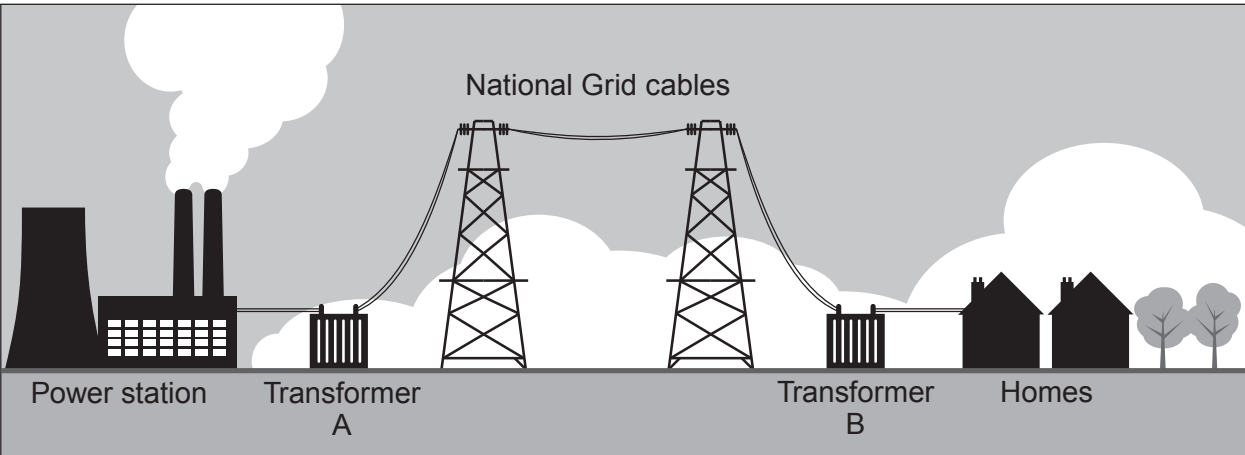
.....

(ii) Suggest **one** change that can be made to the **coil of wire** so that wind speeds less than 0.5 m/s can be measured by the voltmeter. [1]

.....



5. The National Grid includes a large network of cables that connects power stations to homes.



(a) Describe the purpose of transformers A and B in the National Grid.

Include in your answer:

- how electrical energy is transmitted **efficiently**
- how electrical energy is transferred **safely** to homes.

[6 QER]

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Examiner
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- (b) Some houses have solar panels on their roof.
They produce electricity.
Explain why these houses are still connected to the National Grid. [2]

.....

.....

.....

- (c) A housing development of 20 new houses needs to be connected to the National Grid.
Each house requires a **maximum current of 60 A**.

- (i) Calculate the total current needed if all 20 houses are using the maximum current. [1]

Total current = A

- (ii) The mains voltage in the houses is 230 V a.c.

I. Use the equation:

power = voltage × current

to calculate the maximum total power supplied to the 20 houses. [2]

Maximum power = W

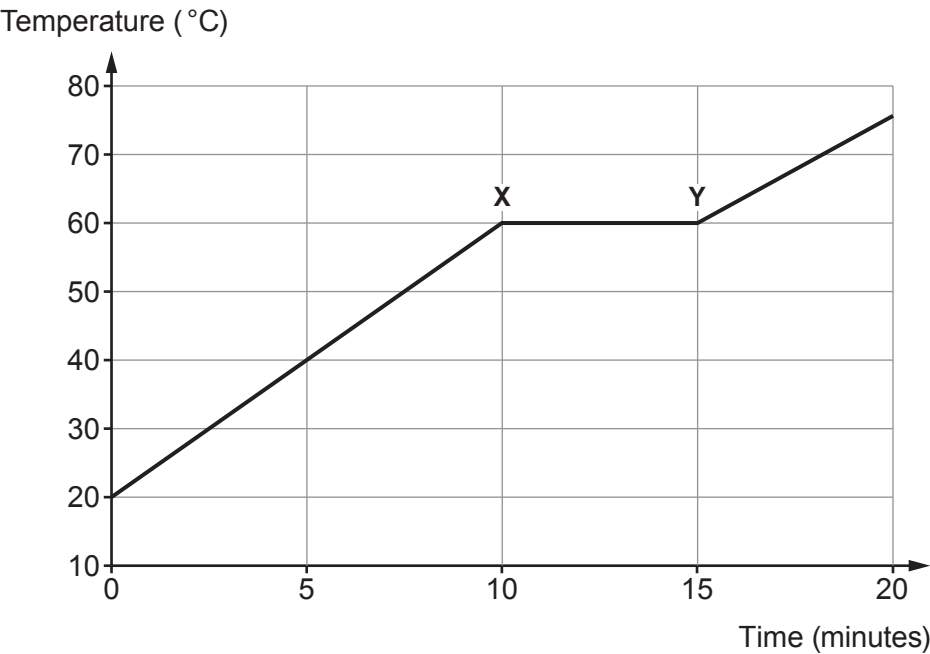
- II. Convert your answer for the maximum power into kW. [1]

Maximum power = kW

12



6. The graph below shows the results obtained when **36 g of solid wax** was heated using a heater. The temperature of the wax was recorded every minute.



(a) Explain why the energy supplied from the heater between **points X and Y** doesn't produce a rise in temperature of the wax. [2]

.....

.....

.....

(b) (i) Use the equation:

thermal energy supplied = mass × specific latent heat

to calculate the amount of thermal energy supplied to the 36 g of wax between **points X and Y**. [2]

(Specific latent heat of fusion (L_f) for wax = 220 J/g)

Thermal energy supplied = J



Examiner
only

(ii) All the energy is transferred from the heater to the wax during the 5 minute interval between **X and Y**.

I. Calculate the time, in seconds, of the interval between **X and Y**. [1]

Time = s

II. Use the equation:

$$\text{power} = \frac{\text{energy transferred}}{\text{time}}$$

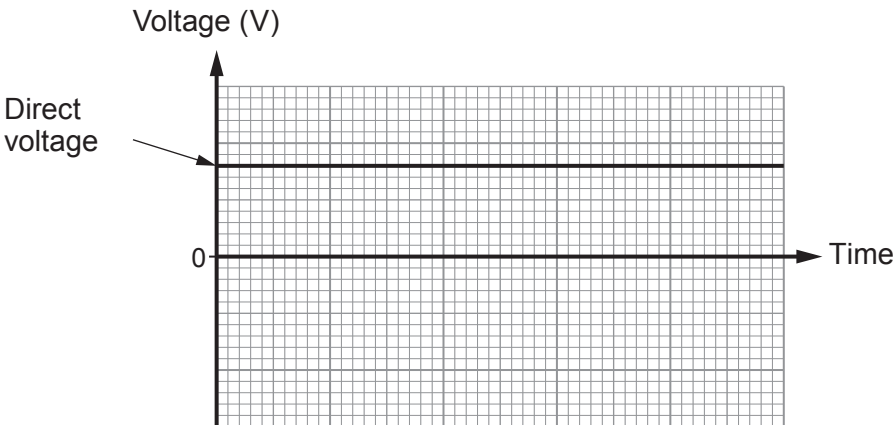
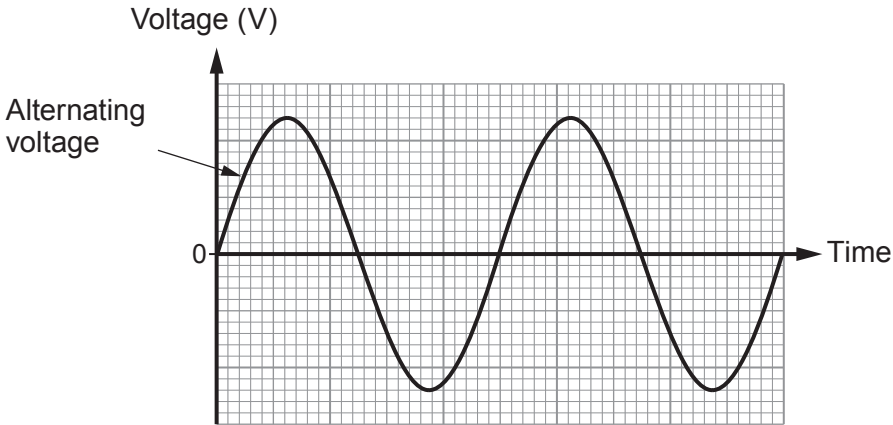
to calculate the power of the heater. [2]

Power of heater = W

7



7. (a) The mains electricity supply to homes is at a voltage of 230 V a.c.
In contrast to this, batteries supply d.c.
These are each shown in the diagrams below.



An alternating voltage changes in size, but a direct voltage keeps the same size.
State **one** other difference between the two.

[1]

.....



(b) Household circuits may be protected by a series of devices. These include:

- a miniature circuit breaker (mcb)
- a fuse in a plug
- a residual current circuit breaker (rccb).

The list below gives three situations in which one or another of these devices stop the power from being supplied.

1. An electric lawn mower cuts through the cable.
2. Too much current flows in a ring main because too many appliances are being used at the same time.
3. The live wire in a kettle touches its neutral wire.

(i) Select and explain one situation above that puts the **user** in danger of an electric shock **and** which one of the devices should be used to keep the user safe. [2]

.....

.....

.....

(ii) State **two** advantages of an mcb over a fuse in a household circuit. [2]

1.
2.

(c) Describe the function of the live and neutral wires in household circuits. [2]

Live wire:

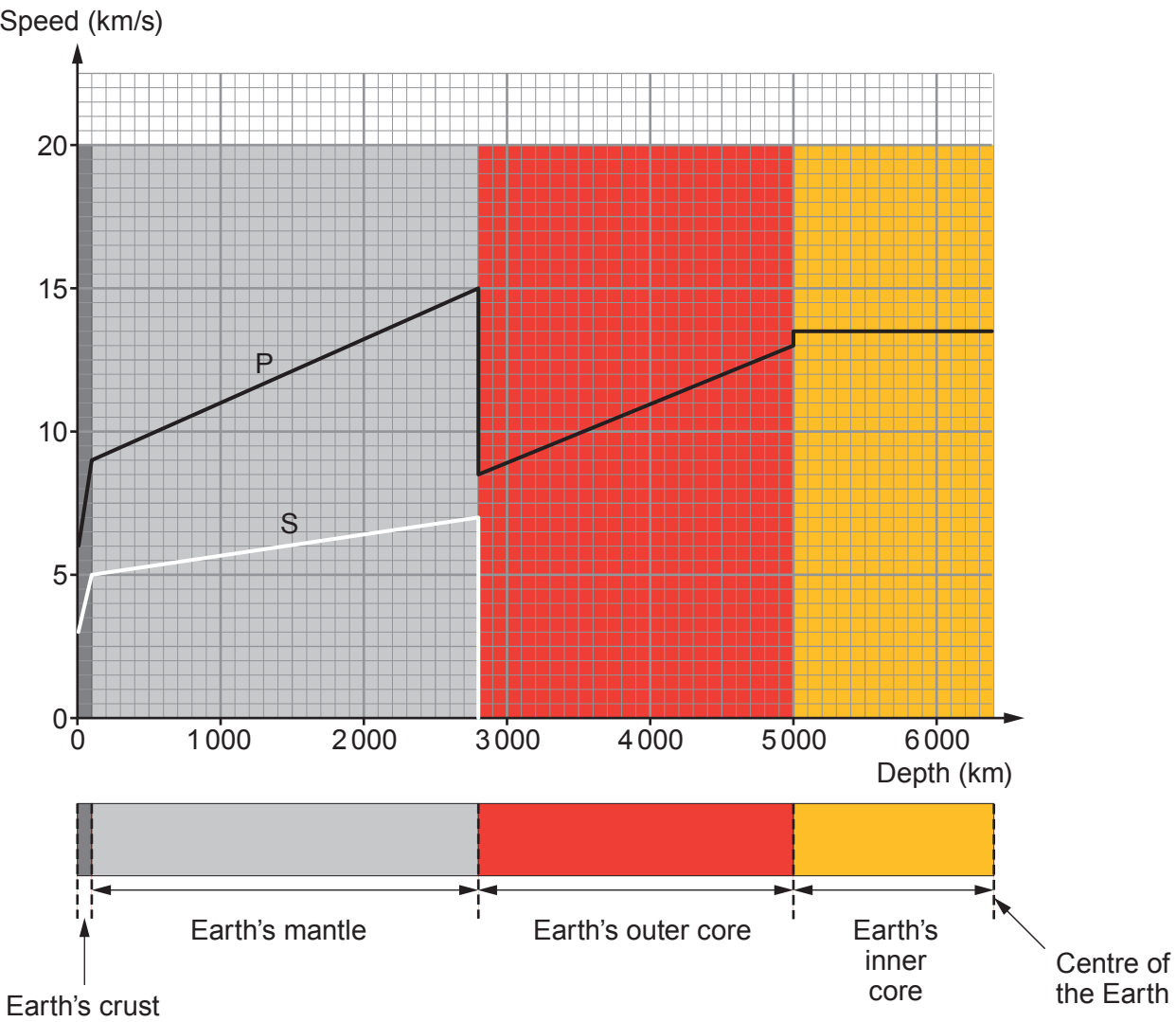
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Neutral wire:

.....



8. P and S waves are produced during an earthquake. They travel at different speeds. The graph shows how the speed of P and S waves change with depth below the Earth's surface.



(a) P waves are longitudinal waves.
Explain what is meant by a longitudinal wave. [2]

(b) Use the graph to calculate the difference in speed between the P wave and the S wave at a depth of **100 km**. [2]

Speed difference = km/s



- (c) (i) Bob claims that the maximum speed of P waves is greater in solids than in liquids. Explain, using data from the graph, whether Bob's claim is correct. [2]

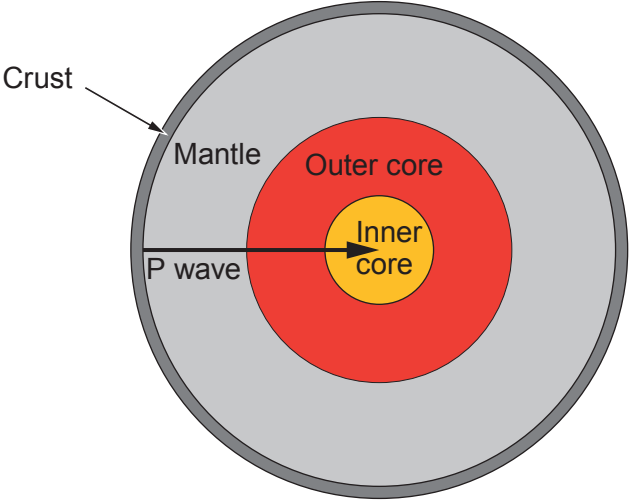
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- (ii) A P wave (shown in Diagram 1) travels a distance of 6 300 km in a straight line to the centre of the Earth in a time of 550 s.

Diagram 1



Bob states that the mean speed of the P wave over this distance is equal to its actual speed only at the depth of 3 500 km. Use an equation from page 2 to show whether Bob's statement is correct. [3]
Space for calculations.

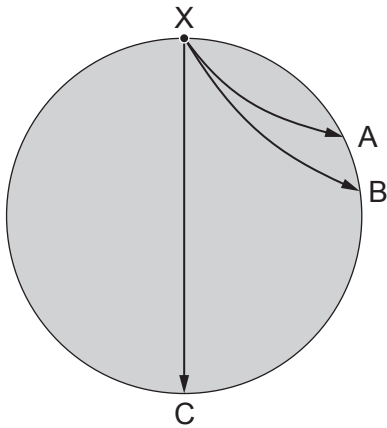
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(d) Diagram 2 shows waves passing through the Earth following an earthquake at point X.

Diagram 2



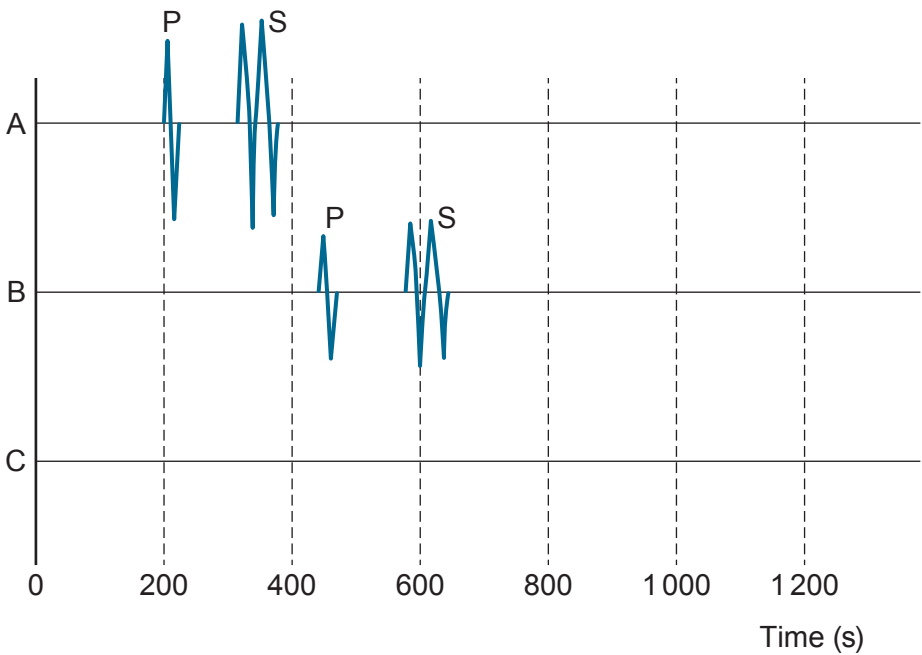
Monitoring stations at A, B and C detect the earthquake by signals that they receive.

- (i) Use the time given in part (c)(ii) to state the time taken for the P wave to travel from **X to C** (ignore the Earth's crust in this). [1]

Time = s

- (ii) The signals received at monitoring stations A and B are shown in Diagram 3. **Add to Diagram 3** the signal that you would expect to be received at station C. [3]

Diagram 3



END OF PAPER



Surname	Centre Number	Candidate Number
First name(s)		0



GCSE

3420U10-1



MONDAY, 19 JUNE 2023 – AFTERNOON

PHYSICS – Unit 1:
Electricity, Energy and Waves
FOUNDATION TIER

1 hour 45 minutes

ADDITIONAL MATERIALS

In addition to this paper you will require a calculator and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen or correction fluid.

You may use pencil for graphs and diagrams only.

Write your name, centre number and candidate number in the spaces at the top of this page.

Answer **all** questions.

Write your answers in the spaces provided in this booklet.

If you run out of space use the additional page at the back of the booklet, taking care to number the question(s) correctly.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.

The assessment of the quality of extended response (QER) will take place in question **5(c)**.

For Examiner's use only		
Question	Maximum Mark	Mark Awarded
1.	12	
2.	7	
3.	4	
4.	4	
5.	12	
6.	9	
7.	12	
8.	7	
9.	13	
Total	80	



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Equations

current = $\frac{\text{voltage}}{\text{resistance}}$	$I = \frac{V}{R}$
total resistance in a series circuit	$R = R_1 + R_2$
energy transferred = power $\times$ time	$E = Pt$
power = voltage $\times$ current	$P = VI$
% efficiency = $\frac{\text{energy [or power] usefully transferred}}{\text{total energy [or power] supplied}} \times 100$	
density = $\frac{\text{mass}}{\text{volume}}$	$\rho = \frac{m}{V}$
units used (kWh) = power (kW) $\times$ time (h) cost = units used $\times$ cost per unit	
wave speed = wavelength $\times$ frequency	$v = \lambda f$
speed = $\frac{\text{distance}}{\text{time}}$	
pressure = $\frac{\text{force}}{\text{area}}$	$p = \frac{F}{A}$
change in thermal energy = mass $\times$ specific heat capacity $\times$ change in temperature	$\Delta Q = mc\Delta\theta$
thermal energy for a change of state = mass $\times$ specific latent heat	$Q = mL$
V_1 = voltage across the primary coil V_2 = voltage across the secondary coil N_1 = number of turns on the primary coil N_2 = number of turns on the secondary coil	$\frac{V_1}{V_2} = \frac{N_1}{N_2}$

SI multipliers

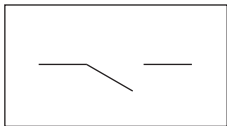
Prefix	Symbol	Conversion factor	Multiplier
milli	m	divide by 1000	1×10^{-3}
centi	c	divide by 100	1×10^{-2}
kilo	k	multiply by 1000	1×10^3
mega	M	multiply by 1 000 000	1×10^6



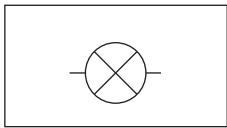
Answer **all** questions.

1. Students are investigating the properties of series and parallel circuits.

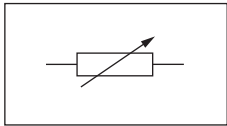
(a) **Draw a straight line** linking the circuit symbol on the left with the name of the component on the right. [3]



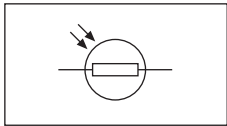
lamp



switch

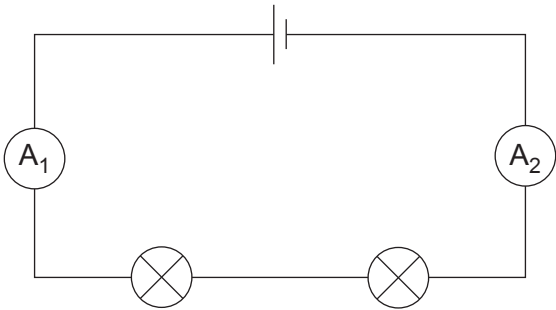


variable resistor



LDR

(b) The students set up the following circuit and measure the current in different places.



(i) Underline words or phrases from the brackets to correctly complete the following sentences. [2]

The circuit is a (**double** / **series** / **parallel**) circuit.

The current reading on ammeter A_2 is (**less than** / **the same as** / **more than**) the reading on ammeter A_1 .



(ii) The battery voltage in the circuit is 3V and the current through the lamps is 0.25A.

I. Use the equation:

$$\text{power} = \text{voltage} \times \text{current}$$

to calculate the power of the circuit. [2]

power = W

II. Use the equation:

$$\text{resistance} = \frac{\text{voltage}}{\text{current}}$$

to calculate the resistance of the circuit. [2]

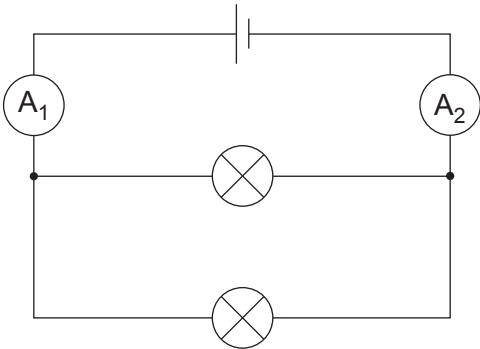
resistance =

III. **Circle** the correct unit for resistance. [1]

J Ω kg



(c) The students now use the **same** components to make the following circuit.



The students notice that the lamps are **brighter**.

Select words or phrases from the box to correctly complete the sentences below.
You may use each word once, more than once or not at all.

[2]

increases	decreases	stays the same
-----------	-----------	----------------

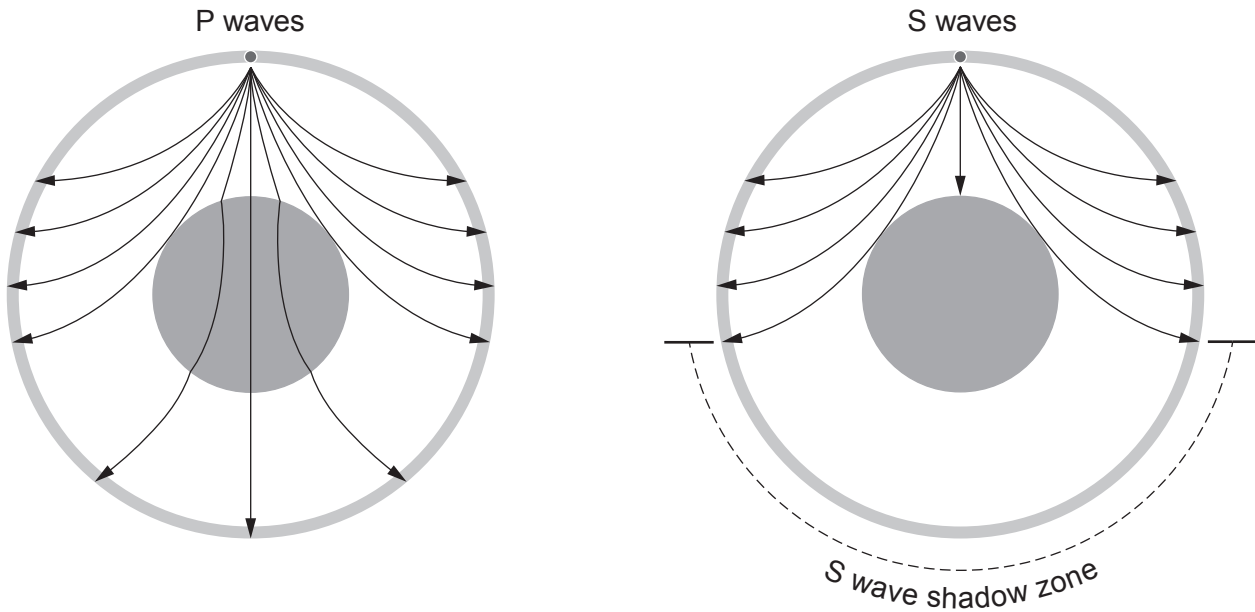
The lamps are brighter because the resistance of the circuit
so the current through each lamp

12

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05



2. The diagram shows the path of seismic P and S waves through the Earth.



(a) (i) Underline the correct word in the bracket. [1]

The Earth has a liquid (**core** / **mantle** / **crust**).

(ii) Tick (✓) the **three** correct statements about P and S waves. [3]

- P waves are transverse waves. ☐
- P waves travel faster than S waves. ☐
- S waves cannot travel in a liquid. ☐
- P waves cannot travel in a solid. ☐
- S waves are transverse waves. ☐
- P waves cannot travel in a liquid. ☐



- (b) (i) P waves from an earthquake, which are travelling at a mean speed of 8 000 m/s, reach a seismic monitoring station 500 s later.

Use the equation:

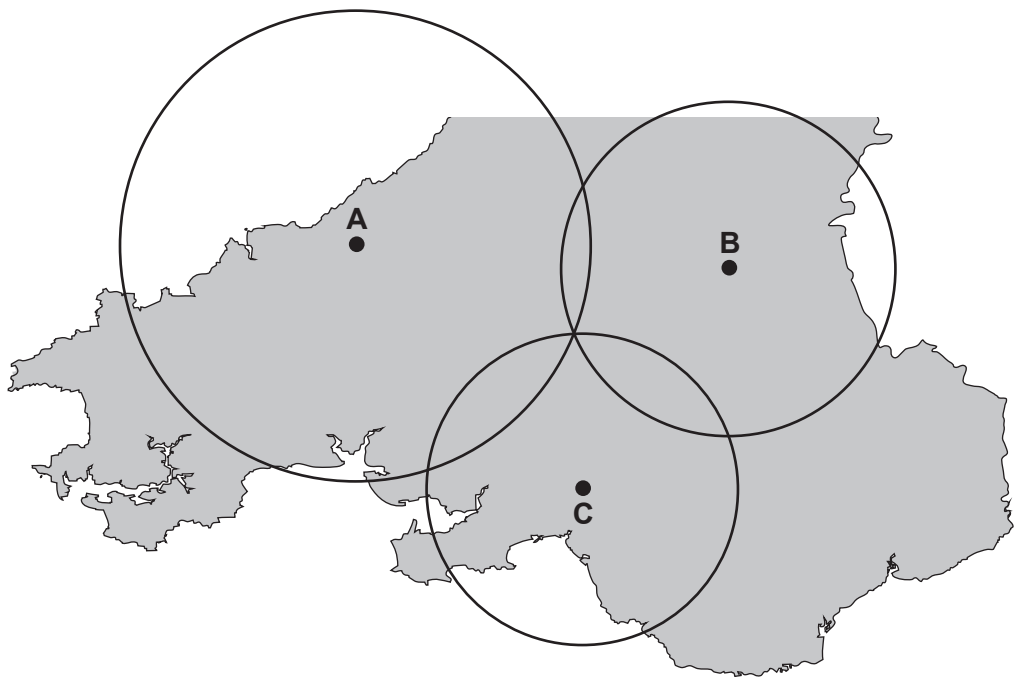
$$\text{distance} = \text{speed} \times \text{time}$$

to calculate the distance travelled by the P waves to the monitoring station. [2]

distance = m

- (ii) **A**, **B** and **C** are monitoring stations in Wales.
They collect data from an earthquake.

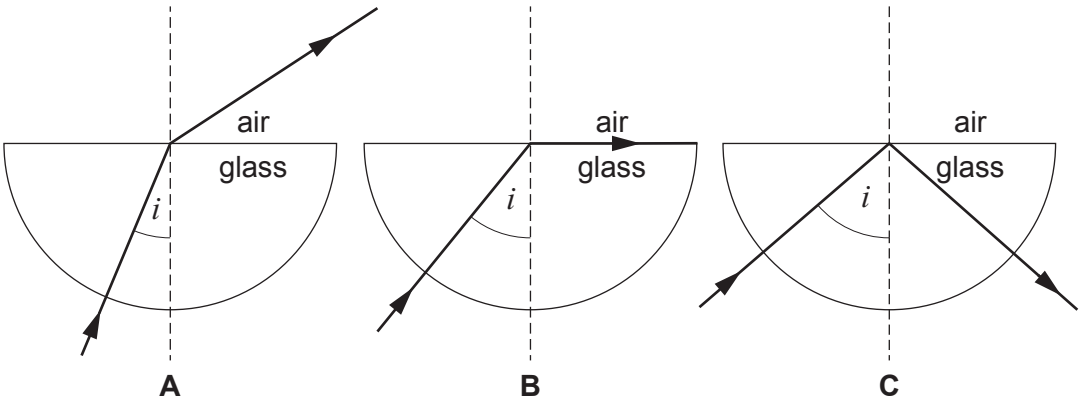
The epicentre is located somewhere on the circles shown in the diagram below.
Mark with a cross (x) the location of the epicentre. [1]



3. A class investigates the total internal reflection (TIR) of light in a glass block.

They change the angle of incidence, i , of a ray of light and draw in the path of the ray out of the block.

Their results are shown in the three diagrams, **A**, **B** and **C** below.



(a) (i) State in which diagram, **A**, **B** or **C**, the angle of incidence is equal to the critical angle. [1]

Diagram

(ii) State which diagram, **A**, **B** or **C**, shows total internal reflection. [1]

Diagram

(b) Tick (✓) the box next to the condition required for total internal reflection to occur. [1]

Light must be travelling from a less dense to a more dense material.

☐

Light must be travelling from one material to a different material of equal density.

☐

Light must be travelling from a more dense to a less dense material.

☐

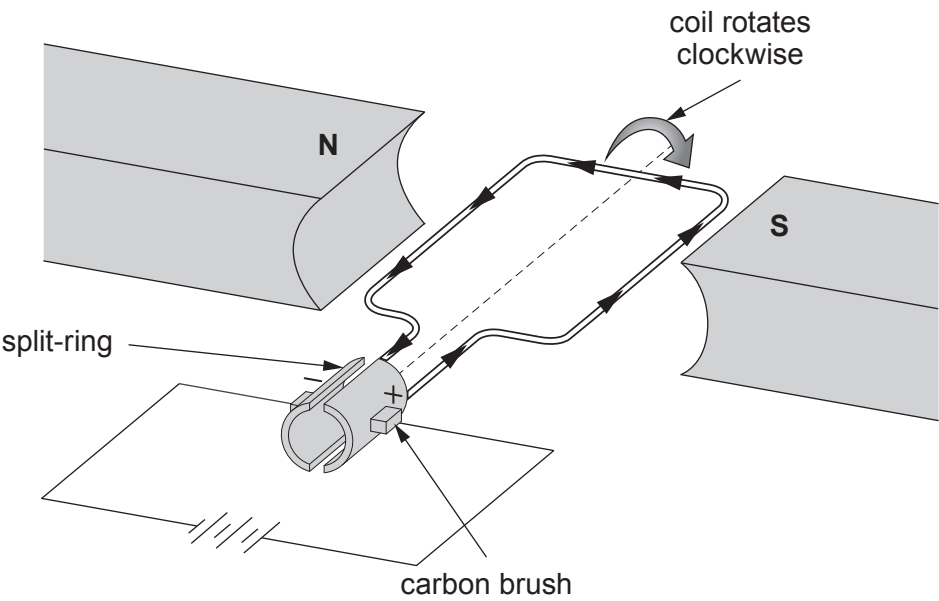
(c) State **one** use of total internal reflection. [1]

.....

4



4. The diagram shows a simple d.c. motor.



The direction of the force on the coil is given by using Fleming's left-hand rule.

(a) **Complete** the labels on the diagram below using words from the box. [2]

CURRENT	MAGNETIC FIELD	FORCE
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Thumb shows the

First finger shows the

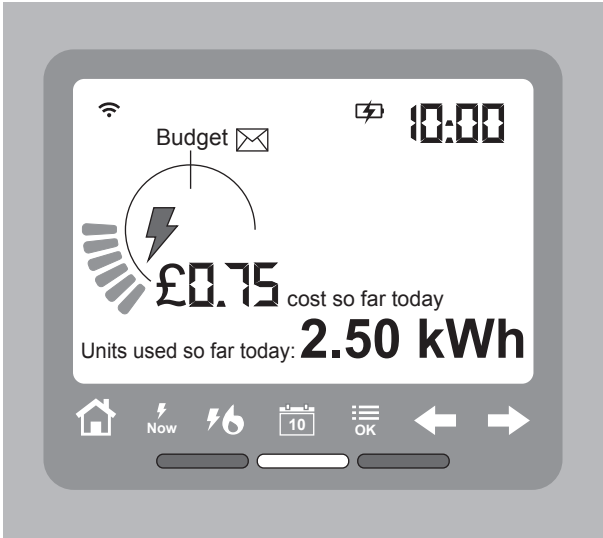
Second finger shows the

(b) State **two** ways of making the motor spin faster. [2]

- 1.
- 2.



5. Smart meters are fitted into some houses to measure how much electricity is used. Householders can monitor and try to reduce their electricity use. The diagram of the smart meter below shows the units used in **10 hours** from midnight to 10.00 am.



- (a) (i) Use the equation:

$$\text{mean power (kW)} = \frac{\text{units used (kWh)}}{\text{time (h)}}$$

and information from the diagram to calculate the mean power of the appliances used in the 10 hours. [2]

mean power = kW



Examiner
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(ii) I. State the cost of the electricity used in 10 hours in pence (p). [1]

cost = p

II. Use the equation:

$$\text{cost of 1 unit (p)} = \frac{\text{cost (p)}}{\text{units used (kWh)}}$$

to calculate the cost of 1 unit of electricity. [2]

cost of 1 unit = p

(b) Give **one** reason, **other than** to save money, why householders should be encouraged to reduce their electricity use. [1]

.....

.....

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(c) Describe the function of an **earth wire**, a **fuse** and a **residual current circuit breaker (rccb)**.

Include in your answer:

- what they are each used to protect
- the fault that causes each of them to work.

[6 QER]

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6. A teacher demonstrates heat transfer by infra-red radiation.

Experiment 1

The teacher places a heater in between a black surface and a silver surface.

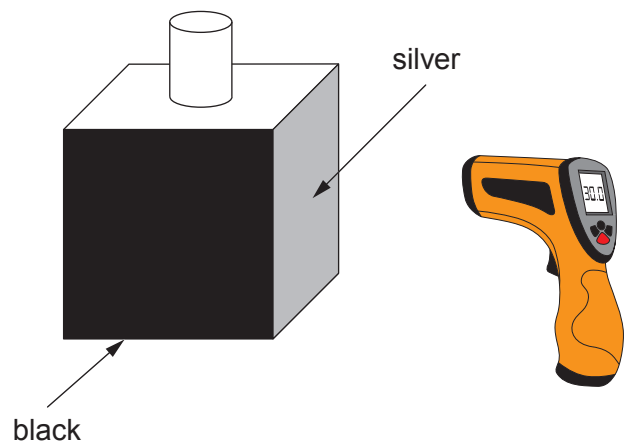
After a few minutes the temperature of each surface is recorded using an infra-red thermometer.



Experiment 2

The teacher fills the container with boiling water.

The infra-red thermometer is then used to take readings next to the side of the container which is black and the side which is silver.



The following results were obtained.

	Temperature (°C)	
	Black surface	Silver surface
Experiment 1	46	32

	Temperature (°C)	
	Black surface	Silver surface
Experiment 2	90	65

- (a) Use the data to complete the following sentences.
- (i) The surface is the better emitter of heat
as shown by the results for **Experiment** because
..... [2]
- (ii) The surface is the better absorber of heat
as shown by the results for **Experiment** because
..... [2]



- (b) Silver coating can be put on the outside of windows to help to keep houses cool in summer and warm in winter.



silver coating

State why the silver coating helps to keep the house warm in the winter. [1]

.....

.....

- (c) A family consider different methods of reducing their energy bills.
Data about these methods are given in the table below.

Method	Cost (£)	Yearly savings (£)
Cavity wall insulation	640	160
Loft insulation	480	160
Silver coating	500	25

- (i) Calculate the payback time for the silver coating. [2]

payback time = years



(ii) The family can only choose to fit one of the items.
leuan suggests that it is better to fit cavity wall insulation than loft insulation.
Use data from the table to explain whether you agree.
Space for calculations.

[2]

Examiner
only

.....

.....

.....

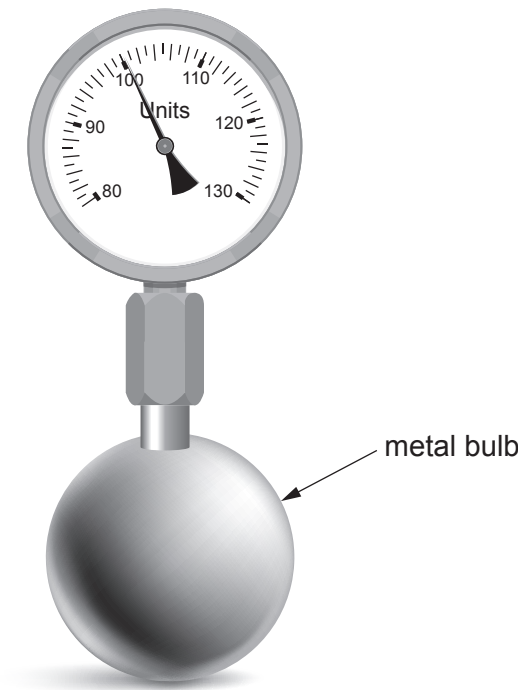
9



7. A teacher uses the apparatus shown below to demonstrate the link between the temperature and pressure of a gas.

The metal bulb, which is attached to a pressure gauge, contains a fixed volume of air.

The bulb is placed in a water bath and measurements of pressure for different temperatures are recorded.



- (a) State the independent variable in this experiment.

[1]

.....

The following results were obtained.

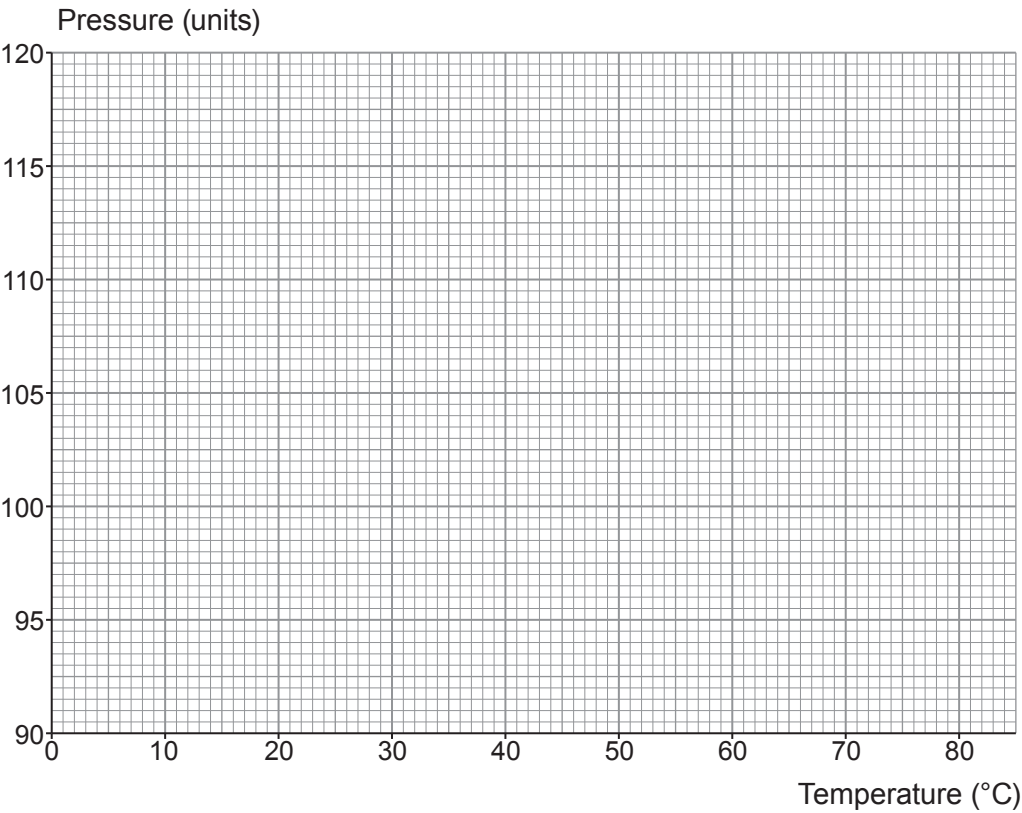
Temperature (°C)	Pressure (units)
0	94
20	100
40	106
60	112
80	118



Examiner
only

(b) (i) Plot the data on the grid below and draw a suitable line.

[3]



(ii) Describe the relationship between the temperature and the pressure of the gas.

[2]

.....

.....



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only

- (iii) I. Use information from the graph or the table to determine the decrease in pressure for every **10 °C** decrease in temperature. [2]

decrease in pressure = units

- II. Use your answer to part I. above to determine the pressure of the gas at **−10 °C**. [1]

pressure = units

- (iv) State the value of absolute zero in °C. [1]

absolute zero = °C

- (c) The specific heat capacity of air, *c*, at constant volume is 720 J/kg °C.

Use the equation:

heat energy supplied = mass × specific heat capacity × change in temperature

$$Q = mc\Delta\theta$$

to calculate the heat energy required to raise the temperature of 0.05 kg of air **from** 20 °C **to** 60 °C. [2]

heat energy = J

12



8. The table below gives information about four types of power station.

The table ranks the power stations in order from 1 to 4 for three different features.
Rank 1 is best and rank 4 is worst.

Power station	Efficiency	Rank	Running cost	Rank	Emissions	Rank
Type A	25 %	4	Second highest	3	Highest polluting emissions	4
Type B		1	Practically zero	1	No emissions	1
Type C	35 %	3	Highest	4	Has cleaner emissions than type A power stations	2
Type D	40 %	2	Second lowest	2	Cleaner emissions than type C power stations but produces radioactive waste	3

(a) Use the information in the table to answer the following questions.

- (i) Gareth says that the best type of power station to recommend overall by ranking is **type B**. Explain whether you agree with him. [2]

.....

.....

.....



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- (ii) The energy sources for different types of power station are **fossil fuel**, **nuclear** and **hydroelectric**.

Complete the table below for the energy sources for types **A**, **B**, **C** and **D**. [3]
Each energy source may be used once, more than once, or not at all.

Type	Energy source
A	
B	
C	
D	

- (b) Use the information below and an equation from page 2 to calculate the % efficiency of a **type B** power station. [2]

Input energy = 200 000 MJ
Heat energy produced = 30 000 MJ
Electrical energy produced = 170 000 MJ

% efficiency =

7



9. Water waves refract when they travel from deep water to shallow water.

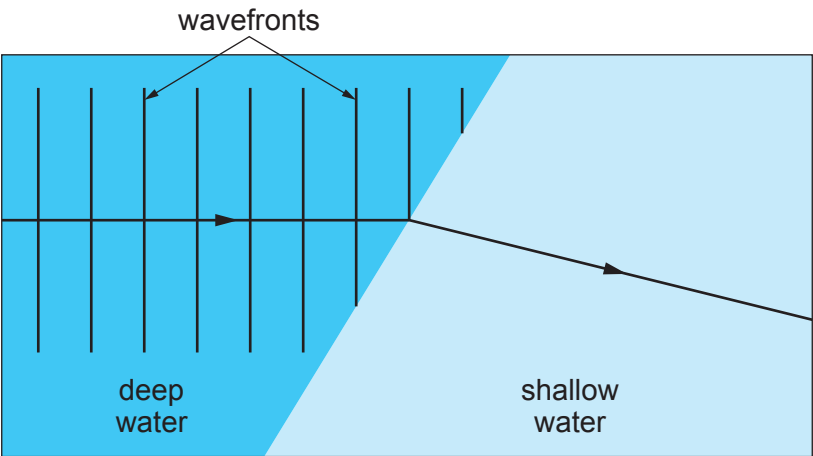
Refraction happens because the speed of the wave changes.

The wavelength also changes because the frequency is constant.

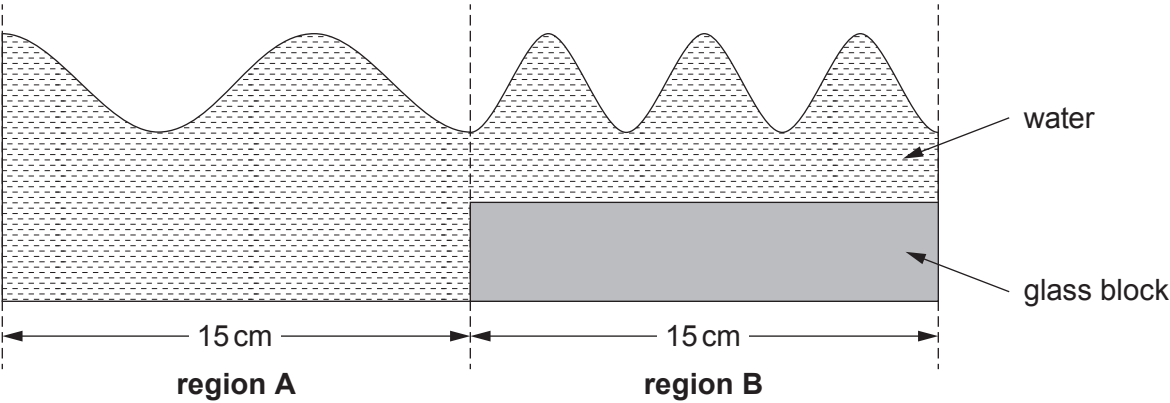
Students use a ripple tank to investigate this effect.

(a) **Complete the diagram below** to show the water waves in shallow water.

[3]



(b) The depth of the shallow water can be changed by using glass blocks of different thicknesses.



Regions **A** and **B** are both 15 cm long.

- (i) I. How many waves are shown in **region A**? [1]
- II. Calculate the wavelength of the waves in **region A**. [1]

wavelength = cm



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(ii) John says that the wave speed in **region B** is greater than the wave speed in **region A**. Explain whether John is correct. [2]

.....

.....

.....

(c) In another experiment using a different tank, students investigate how the depth of water affects wave speed.

They change the depth of the water using different thickness glass blocks.

The water level is kept constant at 10 cm.

The table below shows their results.

Thickness of glass block (cm)	Depth of water (cm)	Wave speed (cm/s)
8	2	60
6	4	75
4		82

(i) **Complete the table.** [1]

(ii) Use the equation:

$$\text{wavelength} = \frac{\text{wave speed}}{\text{frequency}}$$

to calculate the wavelength of water waves of frequency 50 Hz when the thickness of the glass block is 6 cm. [2]

wavelength = cm

TURN OVER FOR THE LAST PART OF THE QUESTION



- (iii) Janet states that when the thickness of the glass block decreases by 2 cm the wave speed increases by a quarter.
Explain to what extent Janet is correct.
Space for calculations.

[3]

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only

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END OF PAPER

